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Greater China Technology Hardware | Asia Pacific

ABF Substrates: Same Thesis, Better Numbers.

We raise our pricing and margin assumptions for ABF substrates. We reiterate OW on Unimicron and NYPCB and raise PT, but also upgrade ZDT to OW, as our latest bottom-up ABF S/D model points to a more severe shortage by 2030.

Key Takeaways

- We estimate the S/D gap will widen to 22% by 2030 (vs 15% previously), driven by increased demand for GPUs, ASICs, and CPUs.
- This will drive stronger pricing and margin tailwinds for ABF substrate suppliers.
- Unimicron PT NT\$1,225 (0.48x/0.24x CY27/28 PEG).
- NYPCB PT NT\$1,275 (0.50x/0.22x CY27/28 PEG).
- Upgrade ZDT to OW with new PT of NT\$570.

Stronger AI demand and more severe undersupply is driving higher pricing and margin assumptions for ABF substrates: We upgraded Unimicron and NYPCB on Feb 22, 2026 ([report here](#)). Over the past three months, both stocks have outperformed the TAIEX, up 123% and 89%, respectively, vs. the TAIEX up 23% over the same period. However, we don't think it's time to take profits, as we continue to see increasing demand, causing further supply constraints and a widening supply/demand gap. Recent checks indicate that ABF undersupply gap would widen to 22% by 2030 (vs 15% previously), which is putting upward pressure on pricing and margins for the substrate suppliers. Thus, we raise our CY26-28 EPS estimates for both stocks, which drives up our PTs for Unimicron to NT\$1,225 (49% upside) and NYPCB to NT\$1,275 (62% upside).

Increasing constraints are also driving a broadening out of the supply ecosystem: As demand gets stronger, we are seeing end-customers choosing to allocate more demand to smaller scale suppliers, as well as customers choosing to work with new entrants. This is why our recent checks indicate that Kinsus (not covered) is working on more different kinds of substrates for Nvidia, and ZDT is also working on qualification for future TPU substrates (in addition to being a key substrate supplier for China AI chips). We believe this is further evidence that customers are concerned about supply tightness and are finding different solutions to ensure business continuity. We upgrade ZDT to OW with new PT of NT\$570 (42% upside).

What are the risks? 1) Weaker demand for PCs, general servers, and AI servers; 2) meaningful capacity expansion plans; 3) T-glass being more constrained than we expect; 4) CoWoP displacing ABF substrates.

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Asia Pacific
Industry View In-Line

Exhibit 1 : What's changed?

Company	Ticker	Rating		PT		Last Price	Upside (%)	EPS change (%)				PEO		
		Old	New	Old	New			26	27	28	29	26	27	28
Unimicron	3023.TW	OW	OW	525	710	62.5	49%	15%	21%	81%	3.13	0.48	0.24	0.24
Nan Ya PCB	8046.TW	OW	OW	1,275	850	785.0	62%	37%	59%	81%	0.39	0.50	0.50	0.22
Zhen Ding	4058.TW	OW	EW	570	250	401.0	42%	18%	38%	38%	0.42	0.51	0.57	0.57

Source: Bloomberg, Morgan Stanley Research estimates. Pricing as of the close on May 15, 2026. .

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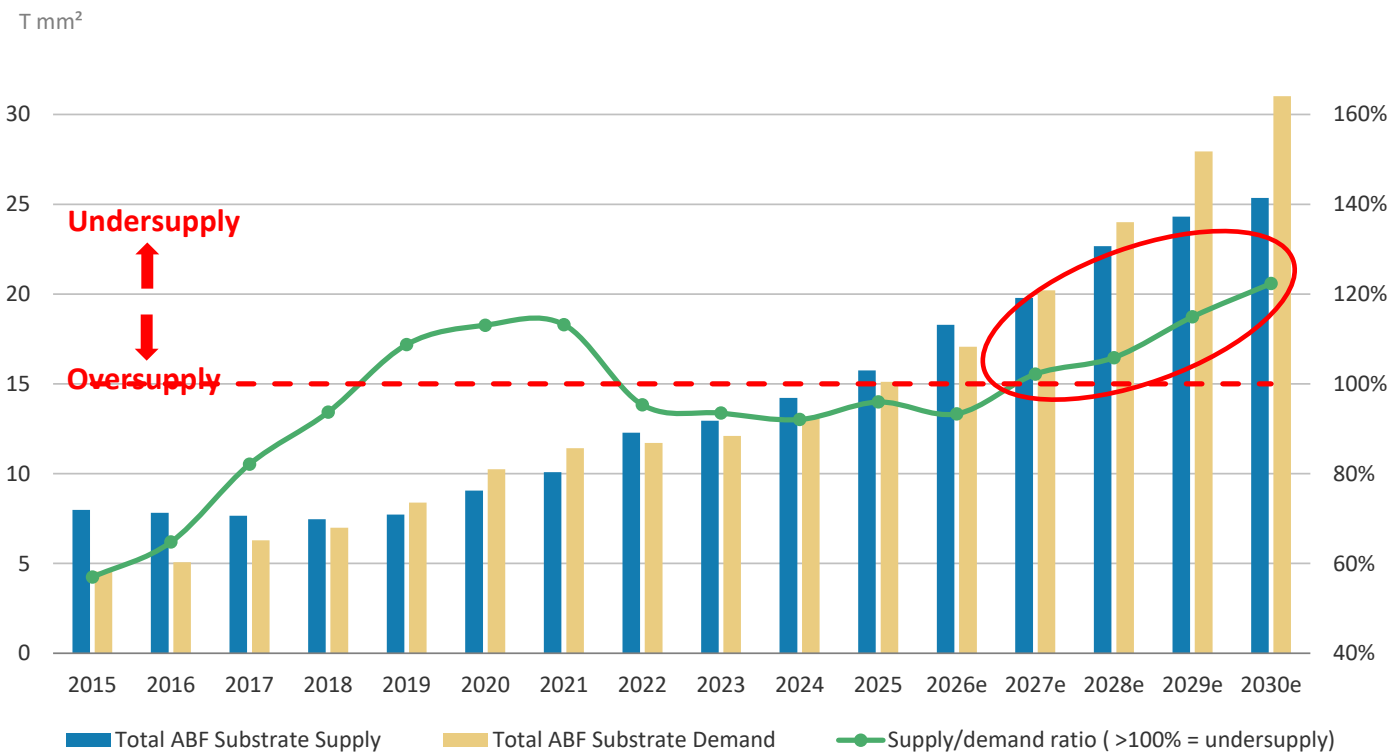
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How Changes to Assumptions Change ABF S/D Dynamics

We now forecast ABF Supply/Demand imbalance to widen by 2030: After baking in latest assumptions and estimates, our bottom-up ABF substrate S/D model points to 22% undersupply by 2030 (vs ~15% previously). Detailed breakdown of the changes in assumptions that resulted in this shortage widening came be found in the next section. But this increased undersupply gives us more confidence on the pricing and margin tailwinds that we expect to see for Unimicron, NYPCB, as well as ZDT.

Why are our supply/demand dynamics largely similar in CY26-28, although demand has increased? This is because we bake in slightly stronger capacity expansion plans for Unimicron. At the beginning of this year, Unimicron only had plans to expand capacity by ~10% y/y, but that has changed in the past few months, and through more de-bottlenecking efforts, Unimicron is able to squeeze out ~40% additional capacity (if we compare YE26e to YE25) across all of its major production sites (four in Taiwan and one in China). Through utilizing all available space and increasing production efficiency, Unimicron is able to quickly add new capacity that can come online this year, compared to any greenfield capacity expansions that would typically take two or more years to bring online.

Exhibit 2: Expecting ABF substrate under-supply gap to widen by 2030 vs our prior estimates



Source: Morgan Stanley Research estimates.

Exhibit 3: Increased demand assumptions is causing CY29-30 under-supply gap to widen vs our prior estimates

ABF SD	2026e	2027e	2028e	2029e	2030e
Old	96.6%	102.6%	106.8%	112.2%	114.6%
New	93.3%	102.2%	105.8%	114.9%	122.3%

Source: Morgan Stanley Research estimates.

Where are we in the cycle?

The ABF substrate market has gone through a few cycles over the past two decades.

We are now still in the early innings of an up-cycle...

There was a prolonged down-cycle from 2012-18 as the PC TAM shrank amid the rise of smartphones and tablets. During this tough period, companies were not able to generate profit and had to compete on price to fill their idle capacity. As a result, during those years, there was no or very limited capacity expansion, and capital expenditure by the major ABF substrate suppliers was kept at a minimum (Exhibit 3).

Even as general server demand picked up, lifting utilization rates across the group going into 2020, ABF substrate suppliers remained cautious on the longer-term demand outlook and stayed disciplined, hesitant to expand capacity. Hence, capex turned up just slightly in 2019-2020, increasing ~US\$600mn vs. ~US\$2.2bn on average per year in 2012-18.

This underinvestment led to a sudden undersupply, as the shock from Covid-19 drove strong work-from-home demand for PCs, as well as increased data usage and storage for general computing driven by online entertainment. This ultimately led to a significant increase in ABF substrate market value between 2020-2022, driven by the sudden surge in demand, along with meaningful pricing adjustments owing to the lack of supply.

During this 2020-2022 period, a lot of capacity expansion was set in motion, as ABF substrate suppliers and customers came to agreement that necessary investments were needed for future growth. Some subsidies were even given to these suppliers to facilitate this capacity expansion. This is why capex jumped from ~US\$2.9bn in 2020 to US\$4.1bn in 2021 and US\$5.5bn in 2022; even in 2023-24 capex remained elevated, as capacity expansion plans continued.

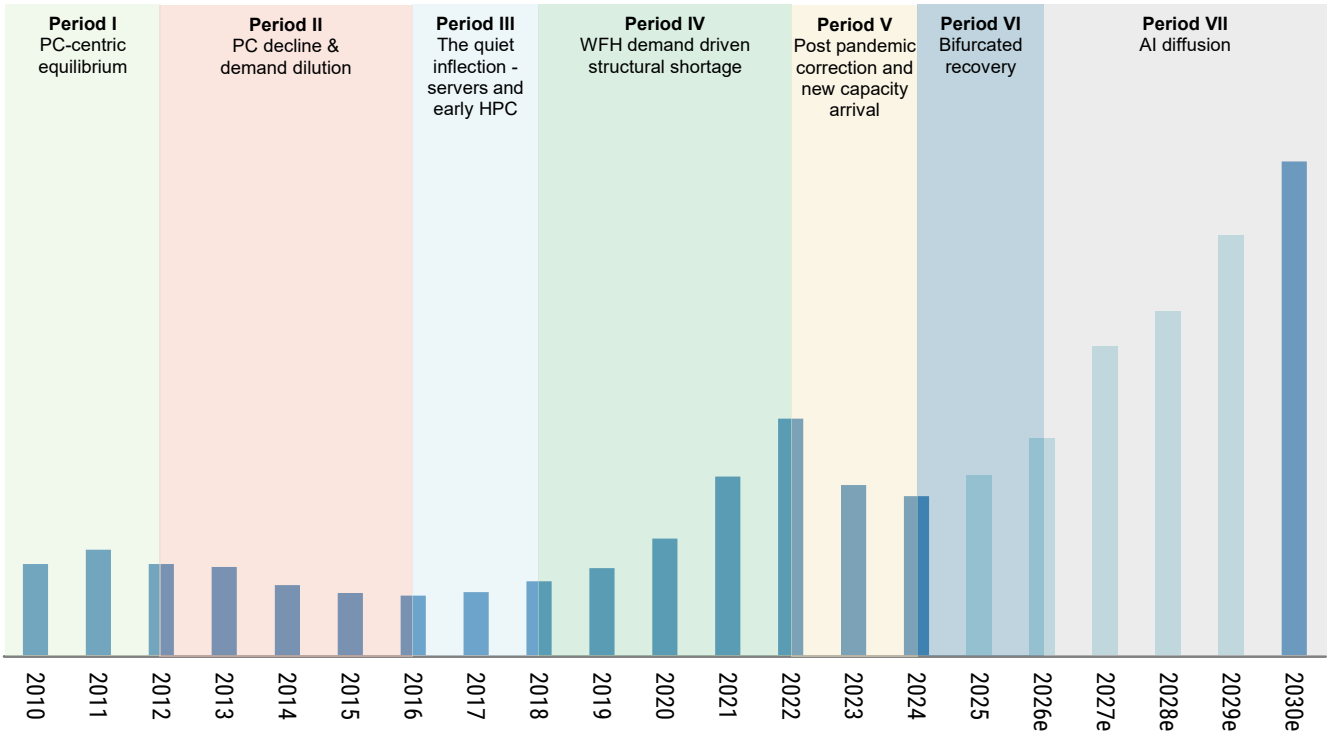
Unfortunately, demand did not play out as expected. As the world came out of the pandemic, demand for PCs and online entertainment fell sharply. This normalization of demand, coupled with capacity expansion from practically every major ABF substrate supplier, resulted in the ABF substrate market once again becoming oversupplied. This led to a sharp decline in ABF substrate market value in 2023-24, as volumes normalized, driving prices down to normalize against the new supply/demand dynamic.

Fortunately, cloud demand remained strong for general computing, and with AI demand starting to pick up, the down-cycle this time around was much shorter (two years) vs. the prior down-cycle (six years).

In 1H24, the ABF substrate market troughed and we started to see an upward inflection since then, but the pace of the recovery has been moderate and bifurcated. Midrange/low-end markets such as PCs have shown moderate demand recovery, with high-end server and AI-related end-demand recovering and growing at a faster pace. This is why we turned the page on our bearish stance in 1H25, but did not turn more bullish until early 2026.

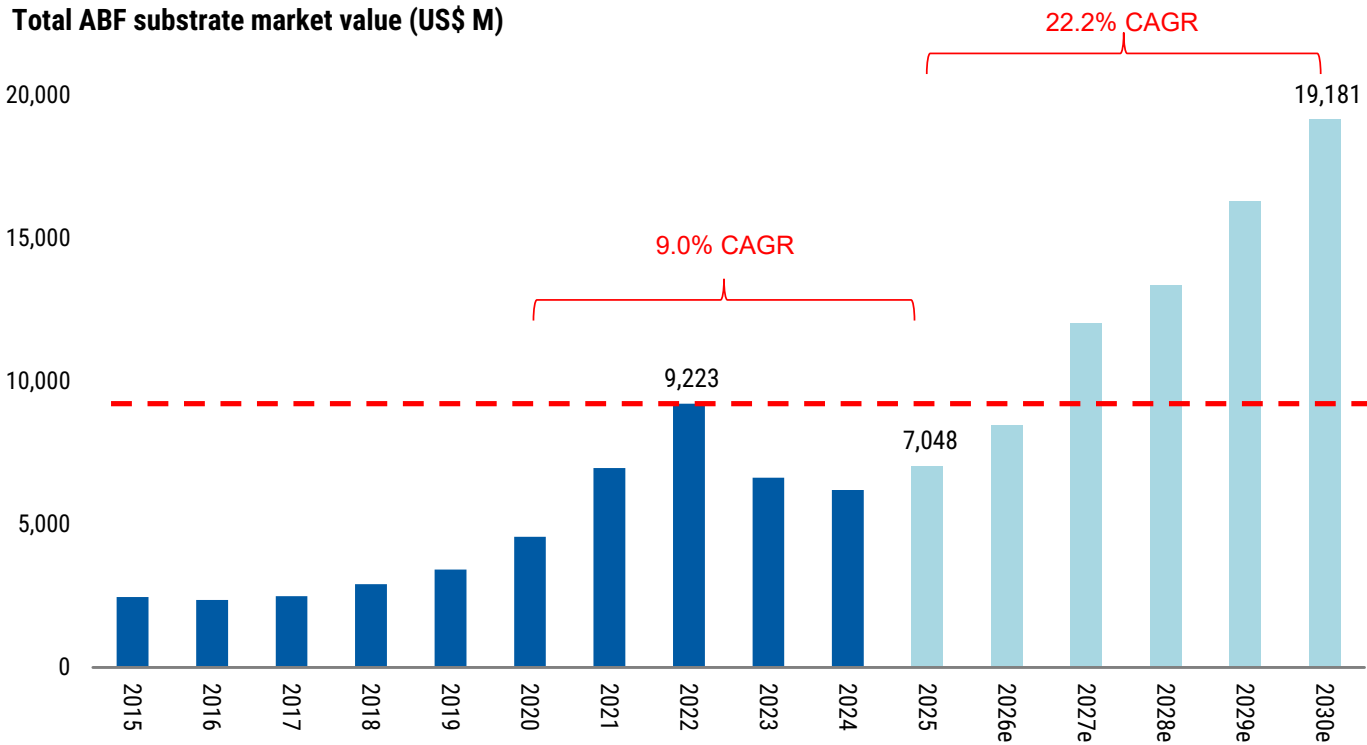
Periods	Description
Period I (2010-2012): PC-centric equilibrium	ABF substrate demand is primarily driven and dominated by CPU and GPU demand for desktops, notebooks, discrete GPUs, with a meaningfully smaller exposure towards server CPUs.
Period II (2012-16): PC decline and demand dilution	Global PC demand entered a structural down-cycle as consumer behavior changed, with smartphones and tablets absorbing more consumer time spent. This resulted in lower volumes and increased pricing competition as suppliers fought to fill idle capacity.
Period III (2016-18): The quiet inflection - servers and early HPC	During these years, the PC market matured, while cloud computing started to take off. At the same time, server CPU substrates quietly grew larger with more layers. We also started to see early accelerators being adopted in high performance computing.
Period IV (2018-2022): WFH demand driven structural shortage	The Covid-19 pandemic drove massive work from home demand for consumer electronics and cloud computing. This sudden surge in PC demand and increased time spent on gaming and online entertainment, bundled with a continuous increase in substrate package size and layer count, resulted in a structural shortage, which also resulted in longer lead-times and raised prices. It was during this period (2021-22) that every ABF supplier globally started to invest in new capacity.
Period V (2022-2024): Post-pandemic correction and new capacity arrival	As the world came out of the Covid-19 pandemic, WFH demand faded and PC and cloud computing demand were negatively affected. But at the same time, the new investments and capacity additions were just starting to slowly come online, resulting in a drop in utilization rates and pricing.
Period VI (2024-2026): Bifurcated recovery	PC demand normalized at the pre-Covid levels, with some replacement demand driven by Windows upgrade, while the recovery and growth of AI and general server demand for cloud picks up at a meaningfully higher growth trajectory.
Period VII (2026-30): AI diffusion	Demand for AI proliferates from training to inference, driving ABF substrate market growth to re-accelerate over the next five years after a two-year down-cycle following the Covid-19 pandemic. This drives increased demand for AI computing, networking, and cloud computing, while demand for PCs remains soft.

Exhibit 4: We are entering the AI diffusion era



Source: Morgan Stanley Research.

Exhibit 5: We now expect ABF substrate demand growth to accelerate, growing at 22.2% CAGR from 2025-2030e (up from 17.9% previously)



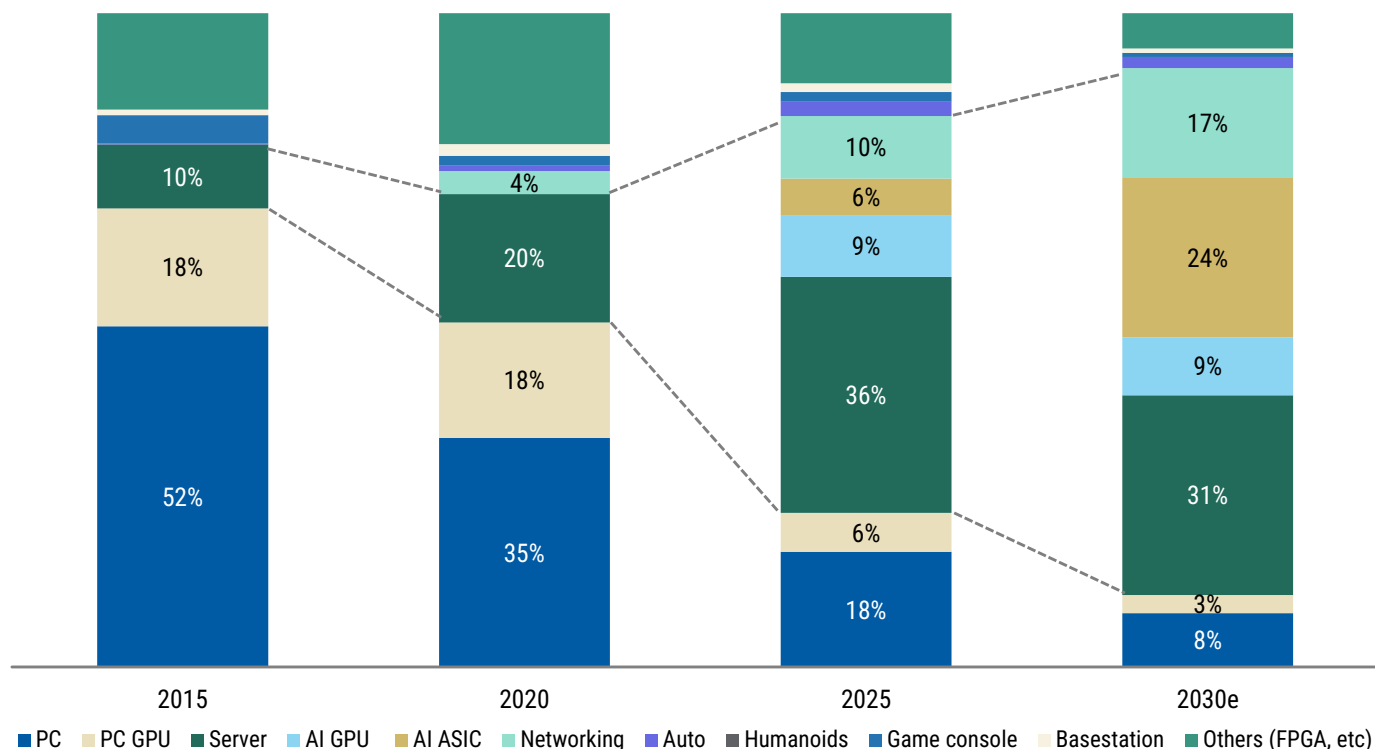
Source: Prismark, Morgan Stanley Research estimates.

Breaking down ABF substrate end-demand

PCs were once the biggest end-market for ABF substrates, in value terms. In 2015 ABF substrates used in relation to the PC market accounted for ~70% of the total ABF substrate value globally, mainly supported by demand for PC CPUs and PC GPUs. But end-demand has shifted meaningfully since then. We estimate that in 2025, ABF substrates used in the PC industry had declined to account for only ~24% of the market, and by 2030, we estimate it will drop further, to less than ~15%.

On the other hand, server, AI GPU, AI ASIC, and networking-related ABF substrate demand is on the rise. In 2015, we estimate that these segments combined were only ~10% of the market, but that rose to ~60% by 2025 and we expect it will account for 75%+ of the entire ABF substrate market value by 2030. In other words, server, AI GPU, AI ASIC, and networking-related applications have been the main growth driver for ABF substrate demand over the past decade and will remain a tailwind to drive even further growth in the coming years. This is the main reason why the ABF substrate market growth will accelerate over the next five years.

Exhibit 6: PC-related ABF demand is declining from ~70% in 2015 to <15% by 2030, while server, AI GPU, AI ASIC, and networking-related ABF demand is rising from ~10% in 2015 to 75%+ by 2030



Source: Morgan Stanley Research estimates.

What does this new S/D dynamic mean for pricing and margins?

We estimate cost inflation will drive ABF substrate pricing to grow 15-20% y/y in CY26, and 20%+ y/y in CY27 mainly driven by an imbalance in ABF substrate S/D. Because our view is that the imbalance will continue to widen in CY28 and thereafter, we are also assuming pricing will continue to move higher in CY28, in the range of 25-40% y/y, depending on the customer. For Unimicron, who has higher exposure to larger customers, we model another 20%+ pricing growth y/y for CY28, but for NYPCB, we estimate pricing growth will accelerate in CY28 to 40%+ y/y growth, given more severe ABF substrate supply shortage.

Any new capacity expansion plans will need at least two years to bring online: So any new meaningful capacity expansion plans that we hear from hereon out will only impact the S/D dynamics for CY29 onwards. So for the next two years, we just need to focus on the changes to end-demand when analyzing this space.

Supply constraints are causing an expansion of the ecosystem to enable more suppliers to move into the high-end

Since the beginning of the year, we have seen more developments happening in the supply chain. We feel like the ABF substrate ecosystem is broadening out to enable more suppliers to prove their capability. We believe this is due to constraints in supply. This is

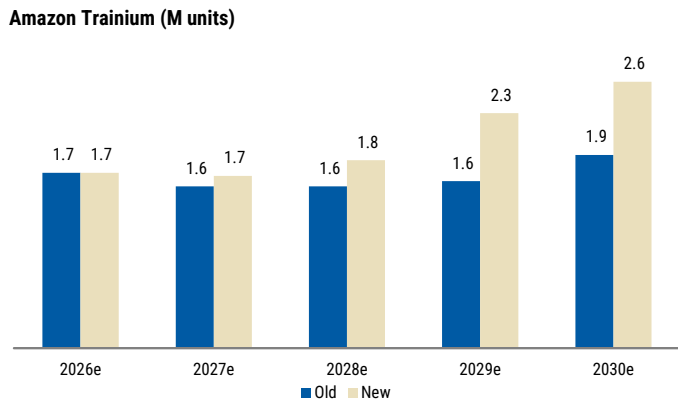
why we see companies like Kinsus (not covered) working on more Nvidia substrate solutions, and even expanding capacity to support more volumes from Nvidia. Additionally, we also see ZDT's utilization rate climbing at its Shenzhen ABF factory, with the company mentioning that it has increased IC substrate revenue 60%+ y/y in 1Q26 and turned profitable for the first time in 1Q. ZDT also mentioned that it sees clear demand visibility for this business and targets to grow by 70%+ y/y for CY26, supported by the ramp-up of its Kaohsiung ABF fab this year for international customers. Moreover, ZDT is also advancing installation of its Shenzhen ABF Fab 2 to support future AI computing-related demand, which we believe is mainly supported by domestic China AI chip demand.

Changes To Our Assumptions

Stronger ASIC demand from hyperscalers

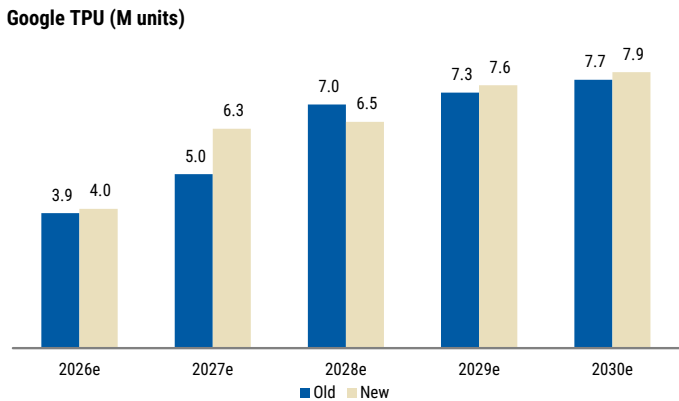
We revisit our ASIC estimates and make revisions based on latest estimates from our semiconductors team. The changes can be seen in the exhibits below, but new estimates show a stronger ramp for Amazon Trainium chips in 2028-2030e, and for Google TPUs, demand is also mostly stronger in over the next five years. Because ASICs are also large body size and high layer count chips, these increased estimates will have a positive impact on ABF substrate demand.

Exhibit 7: Changes to our Amazon Trainium estimates



Source: Morgan Stanley Research estimates.

Exhibit 8: Changes to our Google TPU estimates



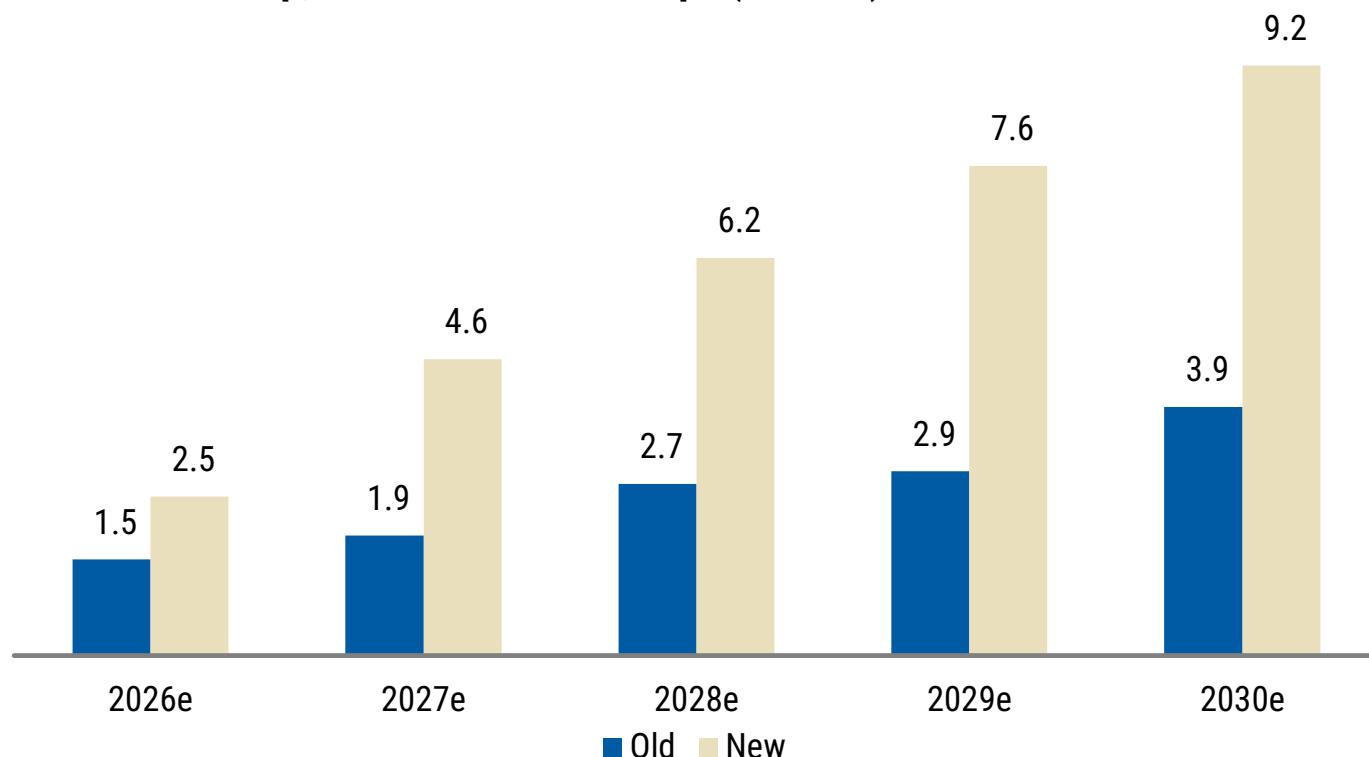
Source: Morgan Stanley Research estimates.

Other ASIC chips

For "Other" ASIC chips, we also take our estimates higher. Included in this section are ASIC chips from Microsoft, other US companies, and China AI chips, such as those from Huawei, Cambricon, Hygon, T-Head, etc. For China AI chips, we included the latest estimates, also from our semiconductor team, which was published in this report ([link](#)). The latest estimates for China AI chips is meaningfully higher than our prior assumption, driving total "Other ASIC chips" category to reach ~9.2M units by 2030, up from ~3.9M units previously.

Exhibit 9: Seeing increased demand from China AI chips

"Other ASIC" chip, includes China AI chips (M units)



Source: Morgan Stanley Research estimates.

The 'agentic economy' is scaling faster than initially anticipated

Please see [here](#) for our [Global Tech team's latest TAM forecast](#). In the report, our base-case total server CPU TAM rises 25% to US\$125B by 2030, driven by a US\$79B orchestration CPU layer on top of US\$45B host/cloud CPU TAM, which drives an incremental ~12-13M units of CPU by 2030. The bull case reaches US\$283B as dedicated agentic CPU racks scale.

From US\$100bn CPU TAM to US\$125bn

In our initial note, [Rise of the AI Agent - Global Implications](#), we parsed the Host CPUs from Orchestration CPUs. We used Mercury data to estimate the Host CPU market (Grace, Vera, Graviton etc) at US\$45bn by 2030. We continue to maintain this assumption in our new model. However, our previous estimates for the orchestration CPU TAM were conservative in particular our assumption for an AI semi TAM of US\$1.2trn, note Nvidia is talking to this being as large as US\$3-4trn. Our previous orchestration TAM of US\$30-60bn implied a CPU TAM of US\$105bn, in-line with Arm's commentary at Arm Everywhere ([link](#)). However, AMD (Advanced Micro Devices) is now talking to a level closer to US\$120bn.

In deriving the Orchestration CPU TAM we take two different approaches: (i) a **bottom-up**

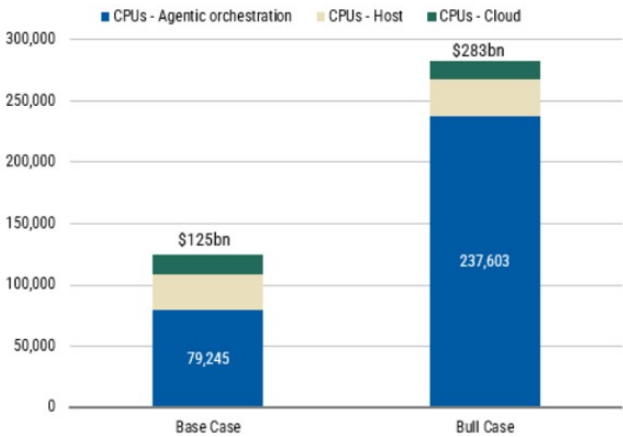
model identifying agents per session, concurrent usage of compute and instructions per token; and (ii) a **top-down** model identifying GW installed per annum, implied racks and the ratio of CPU:GPU.

Exhibit 10: We lift our base case Orchestration CPU TAM to US \$79bn (prev. US\$60bn).

TAM 2030 (\$bn)	Orch.	Host + cloud	Total
Bear Case	32	45	77
Base (current) model	60	45	105
Base (new) Bottom-Up model	79	45	125
Bull Top-Down model	238	45	283

Source: Morgan Stanley Research estimates

Exhibit 11: Our bull case implies a US\$238bn CPU orchestration TAM.

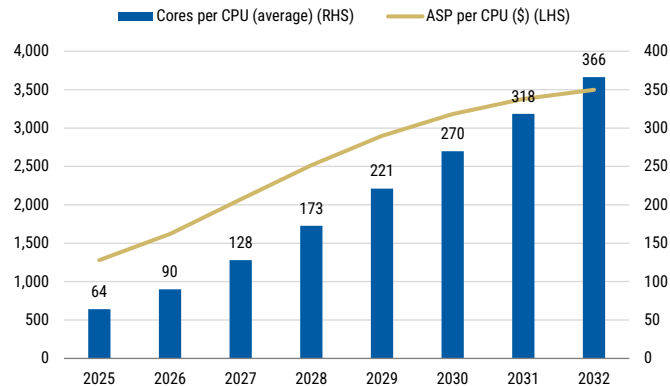


Source: Morgan Stanley Research estimates (e)

Assumptions underlying our model

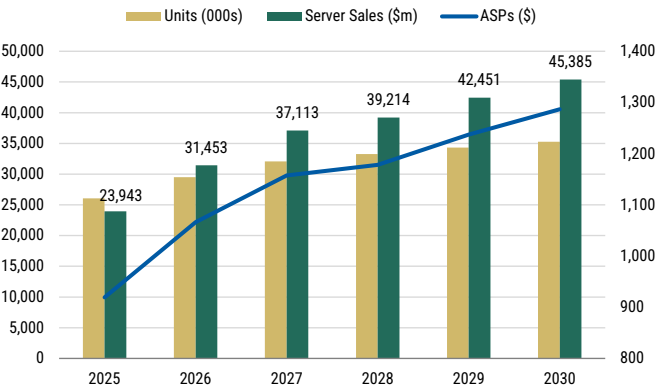
We assume in both our bottom-up and top-down models that a new generation of CPUs per annum implies that pricing per core declines 10% yy. However, in-line with current core trends we expect core counts per CPU to grow from 64 today to c.130 by FY27 and to 200-500+ by 2030. All in, this implies the FY27 ASP per CPU will come in at US \$2,000+ growing to at least US\$3,000 by 2030. For the head node market we model a combination of head node CPUs and regular CPU cloud servers. We expect CPU server sales to grow from US\$31bn FY26 to US\$45bn FY30.

Exhibit 12: We model the ASP per CPU to grow in-line with cores per CPU.



Source: Morgan Stanley Research estimates

Exhibit 13: We continue to expect CPU server sales to grow to \$45bn by 2030.



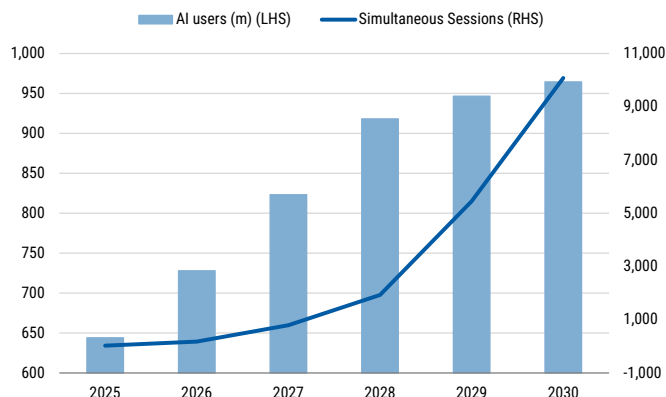
Source: Morgan Stanley Research estimates

A US\$79bn agentic CPU TAM base case...

In deriving our base case we lean on the heavy use of agentic workflows by knowledge workers while assuming that single thread CPUs dominate. In this bottom-up model we

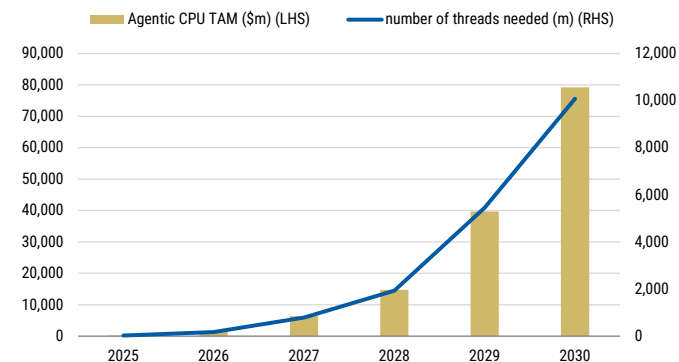
estimate c.1bn knowledge workers globally by 2032 with the geographic mix as 50% Asia, c.25% Europe and 10% to 15% US. We estimate an AI take-up amongst knowledge workers of 78% for FY26 growing to 99% by 2030, while we model the occurrence of concurrent sessions to grow from 4% in FY26 to 19% in 2030. We assume from therein concurrency flattens at 15-20%. While two threads per core is the standard today, we see that reducing on the growing incidence of single threaded CPU cores amidst the need for power efficiency. We model the hyperthreading multiplier to decrease from 2 in 2026 to c.1.5 in 2030 implying the true availability of cores to grow at a 33% CAGR FY25 to FY29. On agentic use today we think today six agents per session is an accurate representation. However we expect this could almost reach 100 agents per session by 2032. We note this may still be conservative as exponential growth would imply trillions of agents per session. Assuming 1 session per thread, this implies 25mn CPUs by 2030. Applying our pricing assumptions this sets our base case orchestration TAM at US\$79bn FY30e. When including our estimate for the Host and Cloud CPU we estimate a US\$125bn CPU TAM.

Exhibit 14: We model simultaneous sessions to grow at a 176% CAGR from FY26 to FY30e.



Source: Morgan Stanley Research estimates

Exhibit 15: Our base case implies a US\$79bn Agentic CPU TAM by 2030.

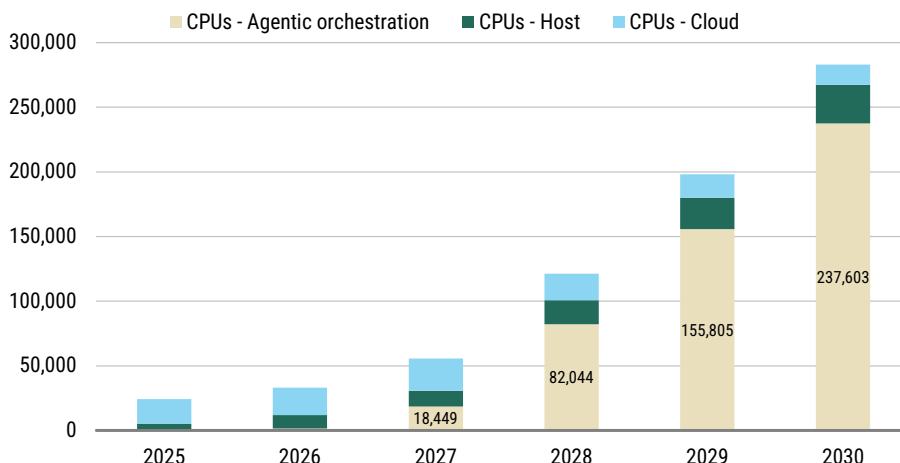


Source: Morgan Stanley Research estimates

... with our bull case implying a US\$238bn agentic CPU TAM.

To derive our bull case we look top-down at what AI data centre installed GW capacity may imply for the CPU:GPU ratio. Our Morgan Stanley global estimates imply AI data centre installed capacity grows from 24GW today before flattening to 35GW in the outer years. Using Nvidia racks as a market proxy we take the power budget per rack to calculate the GPU and CPU power draw per rack and what that would imply for the number of CPUs and GPUs per rack. Assuming pricing for Host CPUs are largely the same as orchestration CPUs this implies a US\$25bn Host CPU TAM. We assume the CPU:GPU ratio ramps from 1:2 to 2:1 by 2030 implying orchestration CPUs grow at a 196% CAGR FY26 to FY30. This sets our orchestration CPU TAM at US\$238bn FY30. Accounting for host and cloud CPUs, our Bull Case CPU TAM is US\$283bn.

Exhibit 16: In our top-down model, we estimate the Agentic CPU TAM to grow at a 251% CAGR from FY26 to FY30.



Source: Morgan Stanley Research estimates

Why are we upgrading ZDT to OW?

Server+Substrate Revenue will grow meaningfully over the next 2-3 years

IC substrate revenue to grow 3x by 2028: According to the company, IC substrate revenue grew 65% y/y in 1Q, which we believe is driven by increased demand and pricing for BT substrates, but also rising demand for ABF substrates. Zhen Ding is a key supplier of ABF substrates in China, utilizing its Shenzhen fab. In fact, the company mentioned that in 1Q26, its Shenzhen ABF fab has finally turned profitable, for the first time. We believe this means that utilization has risen to at least 60%+, and should only continue to grow from here. Our semiconductor team believes that China AI chips will see strong growth over the coming years, growing from ~0.5M units in 2024 to ~7M units by 2030. Our checks indicate that China AI chips use larger substrates, with higher layer count, compared to chips of similar performance from international chip vendors. This is most likely due to the sub-par chip design capabilities. But this becomes a positive for the substrate suppliers, as the pricing of the substrate is determined by the total size in demand. Other than seeing strong growth from China AI chips, we also see ZDT trying to qualify for Google TPU substrate, through MediaTek. If confirmed, this would be more meaningful to the sentiment of the stock than earnings, and help drive a re-rating. Because this would imply that ZDT has the capability to do high-end ASIC substrates, which would also allow ZDT to use that experience and project win to approach other ASIC or GPU customers.

However, in addition to ABF substrates, we also believe ZDT will continue to take share in BT substrates, as we see Unimicron ramping down its BT substrate capacity to focus more on ABF. We believe ZDT will take this opportunity to take more share in BT substrates. Additionally, more demand for memory would also mean more demand for BT substrates

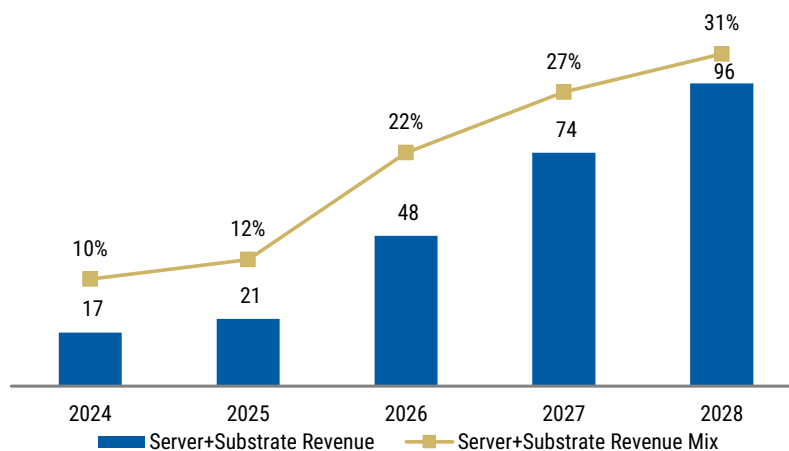
in the coming years.

AI-related PCB will also see strong growth in the coming years: ZDT has already been supplying a lot of general server and AI server peripheral PCBs. It acquired this capability when it acquired BoardTek in 2020. This will continue to grow and even accelerate in the coming years as other Tier1 AI server PCB suppliers are capacity constrained. This gives ZDT more room to expand this portion of its business. Additionally, our checks also indicate ZDT having passed qualification of Nvidia's compute HDI PCB for Rubin. While we believe ZDT's share will remain small, at 5% share, this would still contribute ~NT\$8B revenue annually, and ZDT is able to get its share up to 10%, it mean imply a ~NT\$16B annualized revenue contribution.

Other than this, ZDT is also a significant PCB supplier for optical transceiver modules. For 400G+, HDI PCB is required, and for 800G+ speeds, mSAP capability is needed. ZDT has had mSAP capability for a long-time, being one of the major substrate-like PCB (SLP) for Apple iPhone. This is one of the reasons why ZDT is a significant supplier for optical transceiver module PCBs. We estimate ZDT has ~25% supply share, which would mean that we expect NT\$14-15B revenue contribution this year, growing to ~NT\$25B in CY27. Because there is limited competition today, this business also carries high margin, which we estimate to be at least 40% GM, making it a sizeable profit contributor over the next 2-3 years.

Exhibit 17: We estimate Server+Substrate Revenue Mix would rise to 30%+ by 2028e

Server+Substrate Revenue (NT\$ B) and Mix (%)



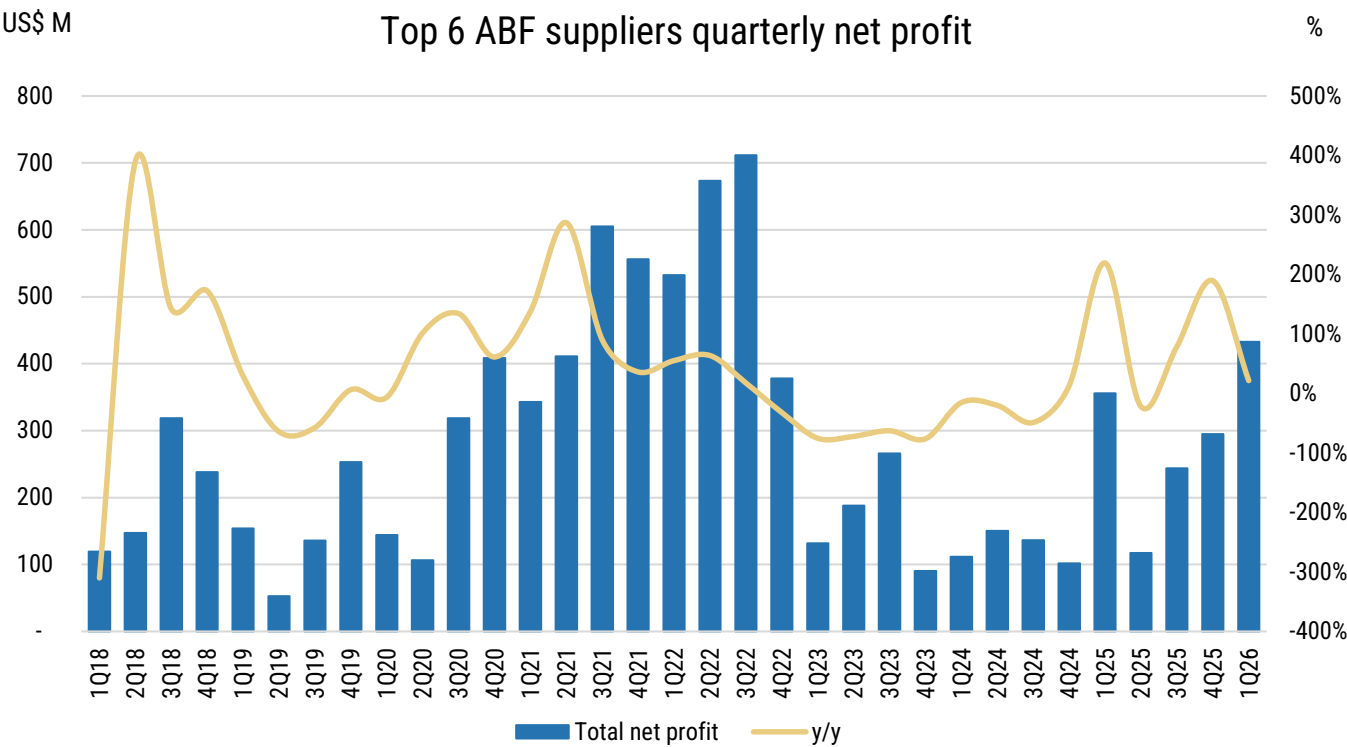
Source: Company data, Morgan Stanley Research estimates.

Our price target for ZDT is NT\$570, which implies 27x CY27e P/E or 20x CY28e P/E. This increase in PT is driven by our CY26-28 EPS raise, but also an increase in medium-term growth rate to 15% (up from 9% previously). The implied CY27e/CY28e P/E multiple for ZDT is lower than Unimicron (51x CY27e P/E, 25x CY28e P/E) or NYPCB (54x CY27e P/E, 25x CY28e P/E), because ZDT does not have as long of a track record on ABF as Unimicron or NYPCB. We think this makes sense, but we can also envision a scenario where ZDT could get re-rated to similar multiples, if ZDT is able to do Google TPU substrates very well over the next 1-2 years and expand its ASIC/GPU customer base.

Profits for the Top 6 ABF Substrate Suppliers Continues to See Improvement

We started to see profit improvements in 2H25, and the recent numbers for 1Q26 continue to see strong q/q growth. We believe very quickly, profitability for the top 6 ABF substrate suppliers will surpass the peak during the Covid years. As we mentioned in the sections above, we believe we are still in the very early innings of this ABF substrate up-cycle, and we believe supply shortage will kick in from CY27 all the way until CY30.

Exhibit 18: Top six ABF suppliers' quarterly net income (1Q18-1Q26)



Source: FactSet, Morgan Stanley Research. The 6 companies include Unimicron, NYPCB, Kinsus, Ibsiden, SEMCO and AT&S.

Exhibit 19: ABF substrate peer comparison table

ABF Substrate Ticker	Company	Closing price (LC) 5/15/2026	Rating	Price target	Market Cap (US\$ mn)	EPS (LC)			Current P/E (X)			PT PEG (X)			P/B (X)			P/S (X)			EV/EBITDA			ROA (%)			ROE (%)		
						26E	27E	28E	26E	27E	28E	26E	27E	28E	26E	27E	28E	26E	27E	28E	26E	27E	28E	26E	27E	28E	26E	27E	28E
3037.TW	Unimicron	821	OW	1,225	39,175	11.62	24.03	49.08	70.7	34.2	16.7	0.42	0.32	0.16	10.4	8.4	6.1	6.6	4.7	3.5	30.8	17.1	9.2	6%	11%	17%	16%	27%	42%
8046.TW	NYPCB	786	OW	1,275	16,108	11.52	23.78	51.05	68.2	33.0	15.4	0.24	0.31	0.13	9.7	8.1	6.0	9.2	6.8	4.6	34.7	20.8	10.4	11%	19%	32%	15%	27%	45%
4958.TW	Zhen Ding	401	OW	570	13,809	13.76	21.11	28.49	29.1	19.0	14.1	0.29	0.36	0.40	2.4	2.2	2.0	1.7	1.4	1.2	10.8	7.7	5.7	5%	7%	9%	8%	12%	15%
4062.T	Ibsiden	15,660	UW	6,500	27,626	180.04	254.29	341.46	87.0	61.6	45.9	3.76	1.49	1.34	7.7	6.9	6.1	8.8	7.3	6.1	27.8	21.7	19.1	5%	6%	8%	10%	12%	15%
009150.KS	SEMCO	1,010,000	OW	920,000	51,101	19,548.81	32,463.68	42,692.45	51.7	31.1	23.7	0.46	0.47	0.75	7.3	6.0	4.8	5.5	4.2	3.4	23.8	16.2	12.4	10%	14%	16%	15%	21%	23%
ATS.AV	AT&S	105	NC	--	4,765	1.32	3.61	8.67	79.8	29.1	12.1	-0.28	0.17	0.09	3.9	3.4	2.4	1.9	1.6	1.4	6.3	6.3	6.3	1%	3%	7%	5%	12%	20%
3189.TW	Kinsus	522	NC	--	8,728	9.28	19.04	32.47	56.3	27.4	16.1	0.33	0.26	0.23	6.4	5.5	4.1	5.4	4.0	2.9	8.5	8.5	8.5	6%	10%	12%	11%	20%	25%
Average						63.3	33.6	20.6	0.7	0.5	0.4	6.8	5.8	4.5	5.6	4.3	3.3	20.4	14.0	10.2	6%	10%	14%	11%	19%	26%			
Median						68.2	31.1	16.1	0.3	0.3	0.2	7.3	6.0	4.8	5.5	4.2	3.4	23.8	16.2	9.2	6%	10%	12%	11%	20%	23%			

Source: FactSet (for AT&S and Kinsus), Morgan Stanley Research estimates (for Unimicron, NYPCB, Zhen Ding, Ibsiden and SEMCO). NC = Not Covered by Morgan Stanley Research.

Unimicron: Estimate Changes

We raise our 2026, 2027 and 2028 EPS estimates by 19%, 21% and 41%, respectively, primarily driven by higher assumptions for ASICs volumes and a bigger agentic AI TAM.

Exhibit 20: Estimate Changes

Year to Dec. 31 (NT\$m)	2026E			2027E			2028E		
	New	Old	% diff	New	Old	% diff	New	Old	% diff
P&L Summary									
Net sales	186,189	169,246	10%	260,849	221,316	18%	352,349	279,329	26%
COGS	(149,522)	(137,245)		(194,590)	(168,945)		(235,051)	(197,521)	
Gross profit	36,667	32,001	15%	66,260	52,372	27%	117,299	81,808	43%
Operating expenses	(17,794)	(13,234)		(19,966)	(13,774)		(22,086)	(14,142)	
Operating income	18,872	18,767	1%	46,293	38,597	20%	95,212	67,666	41%
Non-operating income	5,027	2,280		2,180	2,180		2,180	2,180	
Pre-tax income	23,899	21,046	14%	48,473	40,777	19%	97,392	69,846	39%
Net income	17,879	15,084	19%	36,979	30,665	21%	75,534	53,550	41%
Diluted EPS (NT\$)	11.62	9.80	19%	24.03	19.92	21%	49.08	34.79	41%
Margins									
			<i>ppt</i>			<i>ppt</i>			<i>ppt</i>
Gross margin	19.7%	18.9%	0.8	25.4%	23.7%	1.7	33.3%	29.3%	4.0
Operating margin	10.1%	11.1%	-1.0	17.7%	17.4%	0.3	27.0%	24.2%	2.8
Pretax margin	12.8%	12.4%	0.4	18.6%	18.4%	0.2	27.6%	25.0%	2.6
Net margin	9.6%	8.9%	0.7	14.2%	13.9%	0.3	21.4%	19.2%	2.3

Source: Morgan Stanley Research (E) estimates.

Price Target and Valuation Methodology

Our 12-month price target increases ~56% to NT\$1,225 (from NT\$785), driven by our increased earnings estimates for CY26-28e, our higher medium-term growth assumption and rolling forward the model to CY27e (from CY26e): We believe the company's earnings have returned to a growth track from 2026 onwards and will see stronger earnings growth in 2027, driven by the high-end ABF substrate market's reversion to undersupply and Unimicron's dominant supply share in AI ASICs and also server CPUs, which puts it in a favorable position to benefit from rising AI investments.

We believe the overall ABF substrate market bottomed in 2H24, and growth will accelerate from 2H26 onwards with the ramp of multiple next-generation, large body size ABF substrates for new server GPUs, CPUs, ASICs, and networking chips. New technologies, such as embedded passives and Intel's EMIB-T, will also increase the manufacturing complexity of the package substrates, which adds to the under-capacity as usable output decreases.

Similar to our analysis of other tech hardware companies within our coverage, we use a residual income (RI) valuation model to value Unimicron – we think it derives the most accurate value of the firm as it takes into account cost of equity. Key assumptions:

- Cost of equity of 9.2%;
- Risk-free rate of 1% (10-year Taiwan government note yield);
- Equity risk premium of 8.7%;
- Beta of 1.0;
- **Medium-term growth rate of 15% (from 13% previously)** – we think this is fair given our analysis of the substrate market over the next couple of years;
- Terminal growth rate of 3% (similar to the rest of the companies under our tech hardware coverage).

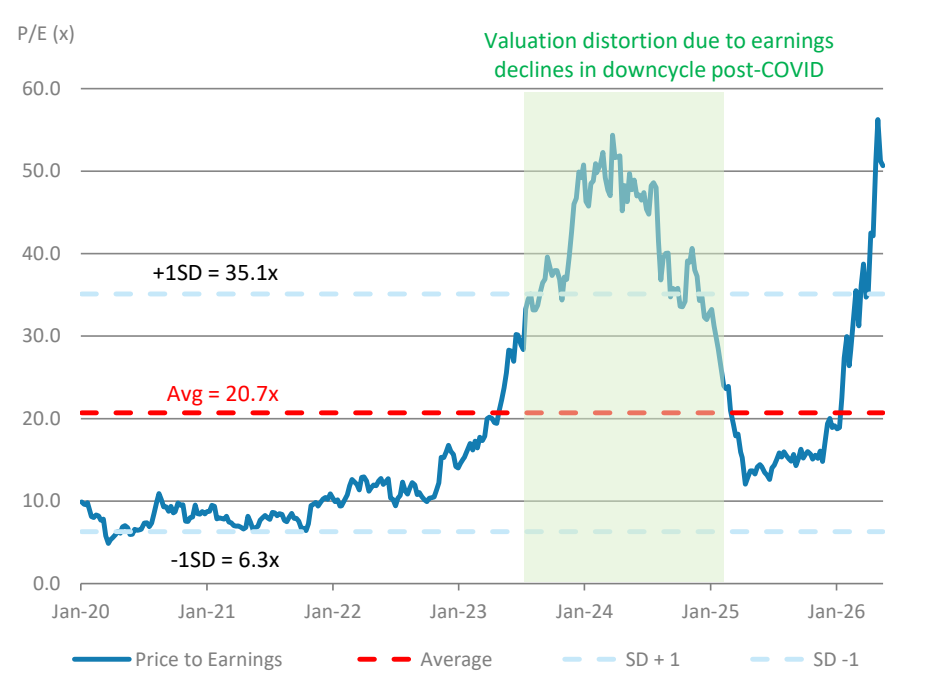
Our bull and bear case values increase by similar magnitudes, to NT\$2,085 and NT\$410, respectively.

Exhibit 21: Unimicron residual income (RI) model

	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E	2035E	2036E	2037E
Total Equity	150,080	208,477	242,619	281,881	327,034	378,958	438,672	507,342	586,314	677,131	753,843
Net Profit	36,979	75,534	86,864	99,894	114,878	132,110	151,926	174,715	200,922	231,061	237,992
Return on Equity	27.2%	42.1%	38.5%	38.1%	37.7%	37.4%	37.2%	36.9%	36.7%	36.6%	33.3%
Beta	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
Equity Risk Premium (Rm-Rf)	8.7%	8.7%	8.7%	8.7%	8.7%	8.7%	8.7%	8.7%	8.7%	8.7%	8.7%
Risk Free Rate (Rf)	1.0%	1.0%	1.0%	1.0%	1.0%	1.0%	1.0%	1.0%	1.0%	1.0%	1.0%
Cost of Equity	9.2%	9.2%	9.2%	9.2%	9.2%	9.2%	9.2%	9.2%	9.2%	9.2%	9.2%
Residual Income	21,933	49,458	61,157	70,160	80,490	92,380	106,062	121,773	139,853	160,643	163,090
Spread	18.1%	33.0%	29.3%	28.9%	28.6%	28.2%	28.0%	27.8%	27.6%	27.4%	24.1%
Beginning Equity Capital	150,080										
PV of Forecast Period	605,147										
PV of Continuing Value	1,130,112										
Equity Value	1,885,339										
No. of Shares	1,539										
Projected Price	1,225.0										

Source: Morgan Stanley Research estimates.

Exhibit 22: Unimicron P/E chart



Source: TEJ, Morgan Stanley Research estimates.

Financials

Exhibit 23: Quarterly Estimates

NT\$ mn	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26E	3Q26E	4Q26E	2025	2026E	2027E	2028E
Sales	30,090	32,466	33,994	34,691	37,446	46,131	50,559	52,052	131,241	186,189	260,849	352,349
Cost of goods sold	-26,063	-28,221	-29,446	-29,219	-30,720	-37,267	-40,359	-41,175	-112,949	-149,522	-194,590	-235,051
Gross profit	4,026	4,246	4,549	5,472	6,726	8,864	10,200	10,877	18,292	36,667	66,260	117,299
Operating expenses	-2,759	-2,736	-3,016	-3,106	-3,969	-4,470	-4,667	-4,688	-11,618	-17,794	-19,966	-22,086
Operating profit	1,267	1,509	1,533	2,366	2,757	4,394	5,533	6,189	6,674	18,872	46,293	95,212
Interest income (loss)	103	62	-7	25	25	25	25	25	183	100	100	100
Investment income (loss)	32	-13	-1	-10	20	20	20	20	8	80	80	80
Gains (Losses) from investment disposal	0	0	0	3	0	0	0	0	3	0	0	0
Gains from FAs disposal	3	337	-29	507	0	0	0	0	817	0	0	0
Exchange gain	452	-1,281	487	349	300	0	0	0	7	300	0	0
Others	-483	-257	964	910	3,197	450	450	450	1,134	4,547	2,000	2,000
Pre-tax profit	1,373	357	2,947	4,150	6,299	4,889	6,027	6,684	8,827	23,899	48,473	97,392
Net Income for Minority	9	231	256	381	338	338	338	338	877	1,351	1,351	1,351
Income tax	-450	-97	-496	-235	-918	-1,222	-1,326	-1,203	-1,277	-4,669	-10,143	-20,507
Net profit	915	30	2,194	3,535	5,043	3,329	4,364	5,143	6,673	17,879	36,979	75,534
EPS (NT\$)	0.59	0.02	1.43	2.30	3.28	2.16	2.84	3.34	4.34	11.62	24.03	49.08
Margins (%)												
Gross margin	13.4%	13.1%	13.4%	15.8%	18.0%	19.2%	20.2%	20.9%	13.9%	19.7%	25.4%	33.3%
Operating margin	4.2%	4.6%	4.5%	6.8%	7.4%	9.5%	10.9%	11.9%	5.1%	10.1%	17.7%	27.0%
Pre-tax margin	4.6%	1.1%	8.7%	12.0%	16.8%	10.6%	11.9%	12.8%	6.7%	12.8%	18.6%	27.6%
Net margin	3.0%	0.1%	6.5%	10.2%	13.5%	7.2%	8.6%	9.9%	5.1%	9.6%	14.2%	21.4%
QoQ Growth												
Sales	2%	8%	5%	2%	8%	23%	10%	3%				
Gross profit	18%	5%	7%	20%	23%	32%	15%	7%				
Operating profit	86%	19%	2%	54%	17%	59%	26%	12%				
Pre-tax profit	331%	-74%	726%	41%	52%	-22%	23%	11%				
Net profit	1531%	-97%	7310%	61%	43%	-34%	31%	18%				
YoY Growth												
Sales	14%	16%	7%	18%	24%	42%	49%	50%	14%	42%	40%	35%
Gross profit	-6%	15%	-8%	61%	67%	109%	124%	99%	12%	100%	81%	77%
Operating profit	-20%	68%	-22%	248%	118%	191%	261%	162%	30%	183%	145%	106%
Pre-tax profit	-59%	-83%	95%	1201%	359%	1270%	105%	61%	21%	171%	103%	101%
Net profit	-62%	-98%	120%	6204%	451%	11144%	99%	46%	31%	168%	107%	104%

Source: Company data, Morgan Stanley Research (E) estimates.

Exhibit 24: Consolidated Financial Summary**Income Statement**

NT\$ mn; FY End Dec	2025	2026E	2027E	2028E
Net sales	131,241	186,189	260,849	352,349
Cost of goods sold	-112,949	-149,522	-194,590	-235,051
Gross profit	18,292	36,667	66,260	117,299
Operating expenses	-11,618	-17,794	-19,966	-22,086
Sales & Marketing	-1,607	-2,534	-3,229	-3,967
G&A	-5,012	-7,806	-8,136	-8,550
R&D	-4,999	-7,454	-8,602	-9,569
Operating profit	6,674	18,872	46,293	95,212
Interest income	183	100	100	100
Investment income	8	80	80	80
Gains from investment disposal	3	0	0	0
Gains from fixed assets disposal	817	0	0	0
Exchange gain	7	300	0	0
Other non-operating income	1,134	4,547	2,000	2,000
Net non-operating profit	2,153	5,027	2,180	2,180
Pre-tax profit	8,827	23,899	48,473	97,392
Net Income for Minority	-877	-1,351	-1,351	-1,351
Income tax	-1,277	-4,669	-10,143	-20,507
Net profit	6,673	17,879	36,979	75,534
Reported EPS (NT\$)	4.34	11.62	24.03	49.08

Balance Sheet

NT\$ mn; FY End Dec	2025	2026E	2027E	2028E
Cash	54,872	82,251	139,598	225,258
Mkt Securities	38	38	38	38
AR/NR	27,570	39,113	54,797	74,018
Inventory	17,802	23,566	30,669	37,046
Others	5,016	7,116	9,969	13,466
Current Assets	105,298	152,084	235,072	349,827
Long-term Investments	836	836	836	836
Fixed Assets	125,866	129,521	121,842	114,440
Other Assets	23,792	23,792	23,792	23,792
Total Assets	255,792	306,233	381,542	488,895
S-t Borrowings	10,845	10,845	10,845	10,845
AP/NP	17,494	23,159	30,140	36,407
Other S-T Liabilities	47,115	62,372	81,171	98,049
Other LT Liabilities	38,457	54,558	76,435	103,247
L-t Debt	34,873	33,872	32,871	31,870
Total Liabilities	148,785	184,806	231,462	280,418
Common Shares	15,296	15,296	15,296	15,296
Other Shareholders' Equity	91,711	106,132	134,784	193,181
Total Equity	107,007	121,428	150,080	208,477
Total Liab./Shrhldr's Equity	255,792	306,233	381,542	488,895

Cash Flow Statement

NT\$ mn; FY End Dec	2025	2026E	2027E	2028E
Cashflow from Operations	14,967	39,199	57,114	88,389
Net Profits	6,673	17,879	36,979	75,534
Depreciation	18,773	19,887	20,076	18,886
Investment losses (income)	0	-80	-80	-80
Investment Disposal Loss	-3	0	0	0
Change in Working Capital	-2,943	3,614	2,993	-2,454
Other adjustments	-7,533	-2,100	-2,853	-3,497
Cashflow from Investing	-23,018	-23,542	-12,397	-11,484
(Purchases) Sale of fixed asset	-24,307	-23,542	-12,397	-11,484
(Purchases) Sale of LT investment	-77	0	0	0
Other adjustments	1,367	0	0	0
Cashflow from financing	19,073	12,022	12,630	8,754
Increase in L-T debt	9,950	-1,001	-1,001	-1,001
Increase in S-T debt	4,331	0	0	0
Issuance of stock	0	0	0	0
Other adjustments	7,087	16,101	21,877	26,812
Exchange Rate Adjustment	96	-300	0	0
Net change in cash	11,118	27,379	57,347	85,659

Financial Ratios

	2025	2026E	2027E	2028E
Margins				
Gross margin	13.9%	19.7%	25.4%	33.3%
Operating expenses/sales	-8.9%	-9.6%	-7.7%	-6.3%
Operating margin	5.1%	10.1%	17.7%	27.0%
Tax rate	14.5%	19.5%	20.9%	21.1%
Pre-tax margin	6.7%	12.8%	18.6%	27.6%
Net Margin	5.1%	9.6%	14.2%	21.4%
YoY Growth				
Turnover	13.8%	41.9%	40.1%	35.1%
Gross Profit	12.1%	100.4%	80.7%	77.0%
Operating Profits	30.4%	182.8%	145.3%	105.7%
Pretax Profits	20.6%	170.7%	102.8%	100.9%
Net Profits	31.3%	167.9%	106.8%	104.3%
Adjusted Cash Dividend (NT\$)	2.00	5.36	11.08	22.64
Net Debt/Equity	-8.6%	-30.9%	-63.9%	-87.6%
Liabilities/Equity	139.0%	152.2%	154.2%	134.5%
Liabilities/Assets	58.2%	60.3%	60.7%	57.4%
ROAE	6.5%	15.7%	27.2%	42.1%
ROAA	2.7%	6.4%	10.8%	17.4%
A/R Turnover (days)	71.2	65.4	65.7	66.7
A/P Turnover (days)	52.7	49.6	50.0	51.7
Inventory Turnover (days)	52.8	50.5	50.9	52.6
Cash conversion (days)	71.3	66.2	66.6	67.6

Source: Company data, Morgan Stanley Research (E) estimates.

NYPCB: Estimate Changes

We raise our 2026, 2027 and 2028 EPS estimates by 37%, 59% and 81%, respectively, primarily driven by higher assumptions for ASICs volumes and a bigger agentic AI TAM.

Exhibit 25: Estimate Changes

NT\$ mn	2026E			2027E			2028E		
	New	Old	Variance	New	Old	Variance	New	Old	Variance
Net sales	55,058	50,539	9%	74,811	61,714	21%	109,356	78,592	39%
COGS	-45,166	-43,127	5%	-55,294	-49,010	13%	-68,495	-55,476	23%
Gross profit	9,891	7,412	33%	19,517	12,704	54%	40,861	23,117	77%
Operating expenses	-1,788	-1,825	-2%	-2,006	-1,968	2%	-2,377	-2,154	10%
Operating income	8,103	5,587	45%	17,511	10,737	63%	38,484	20,963	84%
Non-operating income	811	853	-5%	762	753	1%	762	753	1%
Pre-tax income	8,914	6,440	38%	18,273	11,489	59%	39,245	21,715	81%
Income tax	-1,472	-1,026	43%	-2,905	-1,830	59%	-6,256	-3,464	81%
Net income	7,442	5,414	37%	15,367	9,659	59%	32,989	18,252	81%
EPS (NT\$)	11.52	8.38	37%	23.78	14.95	59%	51.05	28.25	81%
Margins (%)			ppt			ppt			ppt
Gross margin	18.0%	14.7%	3.3	26.1%	20.6%	5.5	37.4%	29.4%	8.0
Operating margin	14.7%	11.1%	3.7	23.4%	17.4%	6.0	35.2%	26.7%	8.5
Pre-tax margin	16.2%	12.7%	3.4	24.4%	18.6%	5.8	35.9%	27.6%	8.3
Net margin	13.5%	10.7%	2.8	20.5%	15.7%	4.9	30.2%	23.2%	6.9

Source: Morgan Stanley Research (E) estimates.

Price Target and Valuation Methodology

Our price target increases ~50% to NT\$1,275, on the back of our raised earnings estimates for CY26-28e and rolling forward the model to CY27e (from CY26e): We believe the company's earnings have returned to a growth track from 2026 onwards and will see stronger growth in 2027, driven by the high-end ABF substrate market's reversion to undersupply, NYPCB's dominant supply share for networking chips, and its proactive pricing strategy, which puts it in a favorable position to benefit from rising AI investments.

We believe the overall ABF substrate market bottomed in 2H24, and growth will accelerate from 2H26 onwards with the ramp of multiple next-generation, large body size ABF substrates for new server GPUs, CPUs, ASICs and networking chips. New technologies, such as embedded passives and Intel's EMIB-T, will also increase the manufacturing complexity of the package substrates, which adds to the under-capacity as usable output decreases.

Similar to our analysis of other tech hardware companies within our coverage, we use a residual income (RI) valuation model to value NYPCB – we think it derives the most accurate value of the firm given that it takes into account cost of equity. Key assumptions (all unchanged):

- Cost of equity of 9.3%;
- Risk-free rate of 1% (10-year Taiwan government note yield);
- Equity risk premium of 8.7%;
- Beta of 1.0;
- Medium-term growth rate of 17% - fair given our analysis of the substrate market over the next couple of years;
- Terminal growth rate of 3% (similar to the rest of the companies under our tech hardware coverage).

Our bull and bear case values increase by similar magnitudes, to NT\$2,280 and NT

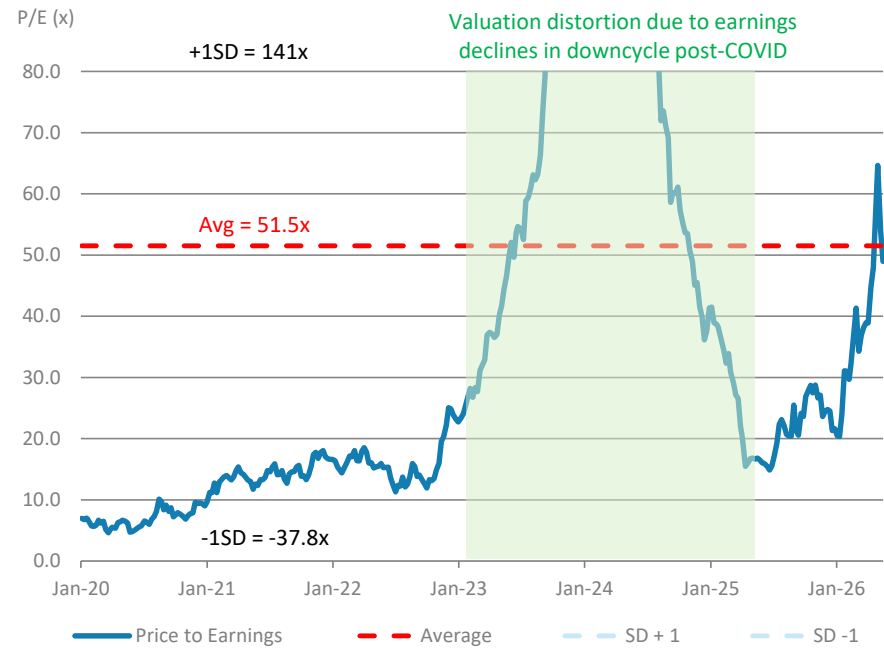
\$650, respectively.

Exhibit 26: NYPCB residual income (RI) model

	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E	2035E	2036E	2037E
Total Equity	62,635	85,297	107,439	133,345	163,656	199,119	240,611	289,157	345,956	412,410	473,944
Net Profit	15,367	32,989	38,597	45,159	52,836	61,818	72,327	84,622	99,008	115,840	119,315
Return on Equity	26.8%	44.6%	40.1%	37.5%	35.6%	34.1%	32.9%	31.9%	31.2%	30.5%	26.9%
Beta	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
Equity Risk Premium (Rm-Rf)	8.7%	8.7%	8.7%	8.7%	8.7%	8.7%	8.7%	8.7%	8.7%	8.7%	8.7%
Risk Free Rate (Rf)	1.0%	1.0%	1.0%	1.0%	1.0%	1.0%	1.0%	1.0%	1.0%	1.0%	1.0%
Cost of Equity	9.3%	9.3%	9.3%	9.3%	9.3%	9.3%	9.3%	9.3%	9.3%	9.3%	9.3%
Residual Income	9,144	22,117	26,261	30,347	35,090	40,614	47,066	54,578	63,366	73,639	72,826
Spread	17.5%	35.4%	30.8%	28.2%	26.3%	24.8%	23.6%	22.7%	21.9%	21.3%	17.7%
Beginning Equity Capital	62,535										
PV of Forecast Period	267,603										
PV of Continuing Value	493,742										
Equity Value	823,880										
No. of Shares	646										
Projected Price	1,275.0										

Source: Morgan Stanley Research estimates.

Exhibit 27: NYPCB P/E chart



Source: TEJ, Morgan Stanley Research estimates.

Financials

Exhibit 28: Quarterly Estimates

	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26E	3Q26E	4Q26E	2025	2026E	2027E	2028E
Net sales	8,458	9,583	10,967	11,165	11,177	13,362	14,734	15,785	40,173	55,058	74,811	109,356
COGS	(8,028)	(8,814)	(9,918)	(9,766)	(9,406)	(11,086)	(11,997)	(12,678)	(36,526)	(45,166)	(55,294)	(68,495)
Gross profit	430	769	1,050	1,398	1,771	2,277	2,737	3,107	3,647	9,891	19,517	40,861
Operating expenses	(397)	(403)	(429)	(435)	(419)	(443)	(459)	(468)	(1,665)	(1,788)	(2,006)	(2,377)
SG&A	(397)	(403)	(429)	(435)	(419)	(443)	(459)	(468)	(1,665)	(1,788)	(2,006)	(2,377)
R&D	0	0	0	0	0	0	0	0	0	0	0	0
Operating income	33	366	620	963	1,352	1,834	2,278	2,639	1,982	8,103	17,511	38,484
Non-operating income	223	(593)	287	446	246	176	193	196	363	811	762	762
Interest income	45	47	19	34	34	34	34	34	144	135	135	135
Investment income	5	(8)	9	12	13	(8)	9	12	18	26	26	26
Disposal of investment	0	0	0	0	0	0	0	0	0	0	0	0
Exchange gain	124	(699)	167	249	112	0	0	0	(158)	112	0	0
Other	49	67	91	151	87	150	150	150	358	537	600	600
Pre-tax income	256	(227)	907	1,409	1,598	2,010	2,471	2,835	2,346	8,914	18,273	39,245
Income tax	(49)	39	(182)	(207)	(290)	(362)	(395)	(425)	(399)	(1,472)	(2,905)	(6,256)
Minority interest	0	0	0	0	0	0	0	0	0	0	0	0
Net income	207	(187)	725	1,202	1,309	1,648	2,075	2,410	1,947	7,442	15,367	32,989
Adj.wtd.avg.shrs (mn)	646	646	646	646	646	646	646	646	646	646	646	646
EPS	0.32	(0.29)	1.12	1.86	2.03	2.55	3.21	3.73	3.01	11.52	23.78	51.05
Fully Diluted shares (mn)	646	646	646	646	646	646	646	646	646	646	646	646
Diluted EPS (NT\$)	0.32	(0.29)	1.12	1.86	2.03	2.55	3.21	3.73	3.01	11.52	23.78	51.05
Margins (%)												
Gross Margin	5.1	8.0	9.6	12.5	15.8	17.0	18.6	19.7	9.1	18.0	26.1	37.4
Operating Margin	0.4	3.8	5.7	8.6	12.1	13.7	15.5	16.7	4.9	14.7	23.4	35.2
Pretax Margin	3.0	(2.4)	8.3	12.6	14.3	15.0	16.8	18.0	5.8	16.2	24.4	35.9
Net Margin	2.5	(2.0)	6.6	10.8	11.7	12.3	14.1	15.3	4.8	13.5	20.5	30.2
QoQ Growth (%)												
Sales	7	13	14	2	0	20	10	7				
Gross Profit	2,441	79	37	33	27	29	20	14				
Operating Profit	NM	10	69	55	40	36	24	16				
Pretax Profit	63	-188	NM	55	13	26	23	15				
Net Profit	16	-190	NM	66	9	26	26	16				
YoY Growth (%)												
Sales	19	18	19	42	32	39	34	41	24	37	36	46
Gross Profit	NM	319	95	8,157	312	196	161	122	920	171	97	109
Operating Profit	NM	NM	396	NM	3,986	401	267	174	NM	309	116	120
Pretax Profit	NM	-226	1,247	794	524	NM	172	101	1,338	280	105	115
Net Profit	NM	-258	1,131	573	531	NM	186	101	856	282	106	115

Source: Company data, Morgan Stanley Research (E) estimates.

Exhibit 29: Consolidated Financial Summary**Consolidated Income Statement**

NT\$mnn (Year End Dec 31)	2025	2026E	2027E	2028E
Net sales	40,173	55,058	74,811	109,356
COGS	(36,526)	(45,166)	(55,294)	(68,495)
Gross profit	3,647	9,891	19,517	40,861
Operating expenses	(1,665)	(1,788)	(2,006)	(2,377)
Operating income	1,982	8,103	17,511	38,484
Non-operating income	363	811	762	762
Interest income	144	135	135	135
Investment income	18	26	26	26
Disposal of investment	-	-	-	-
Exchange gain	(158)	112	-	-
Other	358	537	600	600
Pre-tax income	2,346	8,914	18,273	39,245
Income tax	(399)	(1,472)	(2,905)	(6,256)
Minority interests	-	-	-	-
Net income	1,947	7,442	15,367	32,989
Adj. wtd. Avg. shrs (m)	646	646	646	646
Reported EPS (NT\$)	3.01	11.52	23.78	51.05
Diluted shrs (m)	646	646	646	646
Diluted EPS (NT\$)	3.01	11.52	23.78	51.05

Consolidated Balance Sheet

NT\$mnn (Year End Dec 31)	2025	2026E	2027E	2028E
Cash	8,590	17,495	30,146	54,236
Mkt securities	117	117	117	117
AR/NR	9,552	13,092	17,788	26,002
Inventory	5,068	6,266	7,672	9,503
Others	296	406	552	807
Current Assets	23,623	37,376	56,275	90,665
Long-term investments	500	500	500	500
Fixed assets	35,373	32,258	29,635	27,394
Other assets	1,711	1,711	1,711	1,711
Total Assets	61,207	71,845	88,121	120,271
S/T borrowings	-	-	-	-
AP/NP	1,711	2,116	2,590	3,208
Other ST liabilities	5,478	6,774	8,293	10,272
Total Current Liabilities	7,189	8,889	10,883	13,481
L/T debt	-	-	-	-
Other LT liabilities	7,896	10,821	14,704	21,493
Total Liabilities	15,085	19,711	25,586	34,974
Common shares	6,462	6,462	6,462	6,462
Retained Earnings	13,031	19,042	29,443	52,205
Other SH' Equity	26,630	26,630	26,630	26,630
Total Shareholders' Equity	46,123	52,134	62,535	85,297
Total Liab./SH's Equity	61,207	71,845	88,121	120,271

Consolidated Cash Flow Statement

NT\$mnn (Year End Dec 31)	2025	2026E	2027E	2028E
Operating Cashflow	3,027	10,282	16,570	30,298
Net Profits	1,947	7,442	15,367	32,989
Depreciation & Amort.	6,894	6,013	5,484	5,038
Investment losses/(income)	(18)	(26)	(26)	(26)
Working capital change	12,615	(3,038)	(4,109)	(7,447)
Other adjustments	(18,410)	(110)	(146)	(255)
Investing Cashflow	(1,723)	(2,898)	(2,861)	(2,797)
Capex	(2,422)	(3,000)	(3,000)	(3,000)
Change of L/T investment	0	0	0	0
Change of S/T investment	0	0	0	0
Other adjustments	625	0	0	0
Financing Cashflow	(887)	1,633	(1,058)	(3,412)
Increase in L/T debt	0	0	0	0
Increase in S/T debt	0	0	0	0
Issuance of stock	0	(0)	0	0
Cash dividends	(646)	(1,292)	(4,940)	(10,201)
Other adjustments	(241)	2,925	3,882	6,790
FX adjustment	(309)	(112)	0	0
Net change in cash	109	8,905	12,652	24,090

Consolidated Financial Ratios

	2025	2026E	2027E	2028E
Margins (%)				
Gross margin	9.1	18.0	26.1	37.4
Operating margin	4.9	14.7	23.4	35.2
Pretax margin	5.8	16.2	24.4	35.9
Net margin	4.8	13.5	20.5	30.2
YoY growth (%)				
Sales	24.4	37.1	35.9	46.2
Operating profits	NM	308.7	116.1	119.8
Pretax profits	1,338.4	280.0	105.0	114.8
Net profits	855.6	282.3	106.5	114.7
Others				
Cash dividend payout (%)	66%	66%	66%	66%
Cash div (NT\$)	2.00	7.65	15.79	33.89
Yield (%)	0%	1%	2%	4%
Net Debt/Equity (%)	-19%	-34%	-48%	-64%
Liabilities/Equity (%)	33%	38%	41%	41%
Liabilities/Assets (%)	25%	27%	29%	29%
ROAE (%)	4%	15%	27%	45%
ROAA (%)	3%	11%	19%	32%
AR/NR Turnover (days)	75	75	75	73
Inventory Turnover (days)	46	46	46	46
AP/NP Turnover (days)	16	15	16	15
Cash conversion cycle	105	105	106	103

Source: Company data, Morgan Stanley Research (E) estimates.

ZDT: Estimate Changes

We raise our CY2026, 2027 and 2028 EPS estimates by 18%, 36%, and 38%, respectively. Our earnings increases are driven by our raised demand, pricing, and margin assumptions on ABF substrates, as well as other AI revenue, such as AI server PCBs and optical transceiver PCBs. The differences between the net income changes and the EPS changes are owing to share count change.

Exhibit 30: Estimate Changes

NT\$m	New	Old	Variance	New	Old	Variance	New	Old	Variance
Net sales	220,957	206,610	7%	272,268	236,615	15%	312,343	272,169	15%
COGS	-167,319	-164,180	2%	-201,052	-184,585	9%	-224,876	-208,309	8%
Gross profit	53,638	42,430	26%	71,216	52,031	37%	87,467	63,860	37%
Operating expenses	-29,024	-24,604	18%	-33,545	-27,279	23%	-36,126	-30,387	19%
Operating income	24,614	17,826	38%	37,671	24,752	52%	51,341	33,472	53%
Non-operating income	1,039	2,029	-49%	1,406	1,529	-8%	1,406	1,529	-8%
Pre-tax income	25,652	19,855	29%	39,077	26,281	49%	52,747	35,002	51%
Net income	14,729	11,511	28%	22,596	15,220	48%	30,487	20,275	50%
EPS (NT\$)	13.76	11.71	18%	21.11	15.48	36%	28.49	20.63	38%
Margins (%)			ppt			ppt			ppt
Gross margin	24.3%	20.5%	3.7	26.2%	22.0%	4.2	28.0%	23.5%	4.5
Operating margin	11.1%	8.6%	2.5	13.8%	10.5%	3.4	16.4%	12.3%	4.1
Pre-tax margin	11.6%	9.6%	2.0	14.4%	11.1%	3.2	16.9%	12.9%	4.0
Net margin	6.7%	5.6%	1.1	8.3%	6.4%	1.9	9.8%	7.4%	2.3

Source: Morgan Stanley Research (E) estimates.

Price Target and Valuation Methodology

Our base-case value/price target rises ~128% to NT\$570: It is derived from our multistage residual income (RI) model. We use the company's beginning equity plus the present value of all expected residual income of earnings in excess of the cost of capital.

We use the company's beginning equity plus the present value of all expected residual income of earnings in excess of the cost of capital. The RI is positive when ROAE is above the cost of capital and negative when it is below the cost of capital. We assume a cost of equity of 10%, a medium-term growth rate of 15% (up from 9%) and a terminal growth rate of 3%.

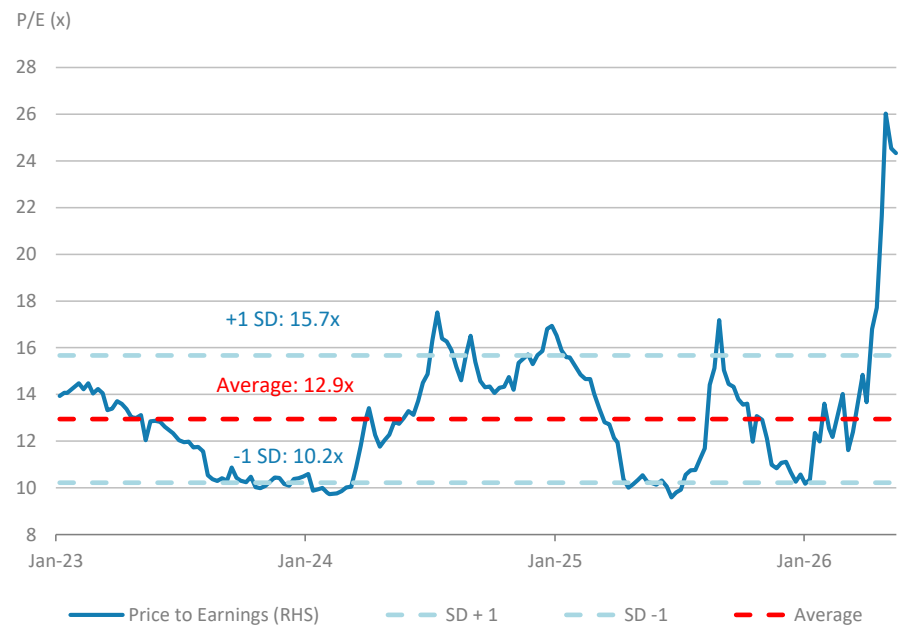
Our bull- and bear-case values also rises by similar magnitudes to NT\$810 and NT\$330, respectively.

Exhibit 31: Residual income (RI) model

NT\$ mn	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E	2035E	2036E
Total Equity	181,802	197,778	218,109	227,357	237,993	250,225	264,291	280,467	299,070	320,463	345,065
Core Net Profit	14,729	22,596	30,487	35,060	40,319	46,367	53,322	61,320	70,518	81,095	93,260
Return on Equity	8.6%	12.4%	15.4%	16.1%	17.7%	19.5%	21.3%	23.2%	25.1%	27.1%	29.1%
Beta (Last 60 Mths)	1.20										
Equity Risk Premium (Rm-Rf)	6%										
Risk Free Rate (Rf)	3%										
Cost of Equity	10%										
Terminal Growth Rate	3%										
Continuing Value Spread	-2%										
Medium-term growth rate	15.0%										
Residual Income	-1,830	4,961	11,302	13,903	18,265	23,281	29,050	35,683	43,312	52,086	62,175
Spread	-1%	3%	6%	6%	8%	10%	12%	14%	15%	17%	19%
Beginning Equity Capital	181,802										
PV of Forecast Period	128,777										
PV of Continuing Value	299,422										
Equity Value	610,001										
No. of Shares	1,070										
Projected Price (EoY)	570.0										
Implied 2027 P/E (x)	27.0										

Source: Morgan Stanley Research estimates.

Exhibit 32: ZDT P/E chart



Source: TEJ, Morgan Stanley Research estimates.

Financials

Exhibit 33: Quarterly Estimates

NT\$ mn	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26E	3Q26E	4Q26E	2025	2026E	2027E	2028E
Sales	40,082	38,203	47,366	56,870	40,728	46,107	63,149	70,973	182,522	220,957	272,268	312,343
COGS	-34,197	-31,193	-36,956	-44,040	-31,917	-35,640	-47,131	-52,631	-146,385	-167,319	-201,052	-224,876
Gross profit	5,885	7,011	10,410	12,830	8,812	10,466	16,018	18,342	36,136	53,638	71,216	87,467
Operating expenses	-4,829	-4,586	-5,880	-6,909	-6,308	-6,911	-7,667	-8,138	-22,205	-29,024	-33,545	-36,126
- Promotion	-486	-493	-630	-744	-616	-697	-859	-869	-2,352	-3,042	-3,666	-4,090
- ADM	-1,781	-1,483	-2,270	-2,512	-2,147	-2,382	-2,610	-2,787	-8,046	-9,926	-11,713	-12,882
- R&D	-2,563	-2,611	-2,981	-3,653	-3,545	-3,831	-4,198	-4,482	-11,807	-16,057	-18,166	-19,154
Operating profit	1,056	2,425	4,530	5,921	2,503	3,556	8,351	10,204	13,932	24,614	37,671	51,341
Non-operating income	401	-153	131	-248	-103	314	414	414	131	1,039	1,406	1,406
Interest income	119	91	49	95	64	64	64	64	355	256	256	256
Investment income	-97	203	242	53	2	100	200	200	401	502	550	550
Disposal of investment	0	0	0	0	0	0	0	0	0	0	0	0
Disposal of fixed assets	16	58	-258	-102	0	0	0	0	-286	0	0	0
Exchange gain	103	-103	0	0	-930	0	0	0	0	-930	0	0
Other	259	-401	98	-294	761	150	150	150	-338	1,211	600	600
Pre-tax profit	1,457	2,272	4,661	5,673	2,400	3,870	8,765	10,618	14,063	25,652	39,077	52,747
Income tax	-431	-885	-1,072	-1,070	-353	-851	-1,648	-1,274	-3,458	-4,126	-6,058	-8,191
Net profit	632	605	2,392	3,161	1,426	1,967	4,735	6,602	6,791	14,729	22,596	30,487
EPS (NT\$)	0.66	0.63	2.46	3.22	1.33	1.84	4.42	6.17	6.91	13.76	21.11	28.49
Margins												
Gross margin	15%	18%	22%	23%	22%	23%	25%	26%	20%	24%	26%	28%
Operating margin	3%	6%	10%	10%	6%	8%	13%	14%	8%	11%	14%	16%
Pre-tax margin	4%	6%	10%	10%	6%	8%	14%	15%	8%	12%	14%	17%
Net margin	2%	2%	5%	6%	4%	4%	7%	9%	4%	7%	8%	10%
QoQ Growth												
Sales	-29%	-5%	24%	20%	-28%	13%	37%	12%				
Gross profit	-49%	19%	48%	23%	-31%	19%	53%	15%				
Operating profit	-81%	130%	87%	31%	-58%	42%	135%	22%				
Pre-tax profit	-81%	56%	105%	22%	-58%	61%	126%	21%				
Net profit	-86%	-4%	295%	32%	-55%	38%	141%	39%				
YoY Growth												
Sales	23%	18%	-6%	1%	2%	21%	33%	25%	6%	21%	23%	15%
Gross profit	10%	65%	-9%	12%	50%	49%	54%	43%	11%	48%	33%	23%
Operating profit	42%	-477%	-24%	7%	137%	47%	84%	72%	20%	77%	53%	36%
Pre-tax profit	-3%	346%	-12%	-27%	65%	70%	88%	87%	-7%	82%	52%	35%
Net profit	-35%	25%	-29%	-28%	125%	225%	98%	109%	-26%	117%	53%	35%

Source: Company data, Morgan Stanley Research (E) estimates.

Exhibit 34: Consolidated Financial Summary**Income Statement**

NT\$ mn	2025	2026E	2027E	2028E
Net sales	182,522	220,957	272,268	312,343
COGS	-146,385	-167,319	-201,052	-224,876
Gross profit	36,136	53,638	71,216	87,467
Operating expenses	-22,205	-29,024	-33,545	-36,126
- Promotion	-2,352	-3,042	-3,666	-4,090
- ADM	-8,046	-9,926	-11,713	-12,882
- R&D	-11,807	-16,057	-18,166	-19,154
Operating income	13,932	24,614	37,671	51,341
Non-operating income	131	1,039	1,406	1,406
Interest income	355	256	256	256
Investment income	401	502	550	550
Disposal of investment	0	0	0	0
Disposal of fixed assets	-286	0	0	0
Exchange gain	0	-930	0	0
Other	-338	1,211	600	600
Pre-tax income	14,063	25,652	39,077	52,747
Income tax	-7,272	-10,923	-16,481	-22,260
Net income	6,791	14,729	22,596	30,487
EPS (NT\$)	6.91	13.76	21.11	28.49

Balance Sheet

NT\$ mn	2025	2026E	2027E	2028E
Cash	71,116	61,515	71,281	83,981
Mkt securities	36	36	36	36
Accounts/Notes receivables	31,412	38,027	46,858	53,755
Inventory	19,616	22,422	26,942	30,134
Other current assets	5,460	5,460	5,460	5,460
Current Assets	127,641	127,460	150,577	173,366
Long-term investments	9,866	10,367	10,917	11,467
Fixed assets	123,737	145,693	148,605	152,383
Other assets	18,799	18,799	18,799	18,799
Total Assets	280,043	302,319	328,898	356,016
S/T borrowings	23,839	23,839	23,839	23,839
AP/NP	23,501	26,862	32,278	36,103
Other ST liabilities	25,481	33,307	38,494	41,456
Other liabilities	21,151	21,151	21,151	21,151
L/T debt	15,359	15,359	15,359	15,359
Total Liabilities	109,330	120,517	131,120	137,907
Common shares	10,706	10,706	10,706	10,706
Capital Collection	0	0	0	0
APIC	52,902	52,902	52,902	52,902
Retained earnings	61,549	72,586	88,562	108,893
Treasury Stock	-52	0	0	0
Other shareholders' equity	45607	45607	45607	45607
Shareholders' equity	170,713	181,802	197,778	218,109
Total Liab./Shrhldr's Equity	280,043	302,319	328,898	356,016

Cash Flow Statement

NT\$ mn	2025	2026E	2027E	2028E
Cashflow from operations	27,617	24,038	26,386	32,856
Net Profits	10,605	14,729	22,596	30,487
Depreciation	18,552	8,044	7,088	6,222
Equity investment losses (income)	0	-502	-550	-550
Other adjustments	-1,541	1,767	-2,748	-3,303
Cashflow from investing	-31,960	-30,000	-10,000	-10,000
(Purchases) sale of FA (capex)	-32,673	-30,000	-10,000	-10,000
(Purchases) sale of L/T investment	342	0	0	0
(Purchases) sale of S/T investment	-601	0	0	0
Other adjustments	972	0	0	0
Cashflow from financing	-1,865	-3,640	-6,620	-10,156
Increase in L/T debt	6,344	0	0	0
Increase in S/T debt	2,191	0	0	0
Issuance of stock	730	0	0	0
Cash dividends	-7,221	-3,692	-6,620	-10,156
Dir. & Emp. Bonus	0	0	0	0
Changes in Treasury Stocks	288	52	0	0
Other adjustments	-4,196	0	0	0
Exchange rate adjustment	-2,177	0	0	0
Net change in cash	-8,386	-9,601	9,766	12,700

Financial Ratios

	2025	2026E	2027E	2028E
Margins				
Gross margin	19.8%	24.3%	26.2%	28.0%
Operating margin	7.6%	11.1%	13.8%	16.4%
Pretax margin	7.7%	11.6%	14.4%	16.9%
Net margin	3.7%	6.7%	8.3%	9.8%
YoY growth				
Sales	6.3%	21.1%	23.2%	14.7%
Operating profits	20.2%	76.7%	53.0%	36.3%
Pretax profits	-6.5%	82.4%	52.3%	35.0%
Net profits	-26.0%	116.9%	53.4%	34.9%
EPS	-28.6%	99.2%	53.4%	34.9%
Net Debt/Equity (Net of mkt secs.)	-19%	-12%	-16%	-21%
Net Debt/Equity	-19%	-12%	-16%	-21%
Liabilities/Equity	64%	66%	66%	63%
Liabilities/Assets	39%	40%	40%	39%
ROAE	4.2%	8.4%	11.9%	14.7%
ROAA	2.5%	5.1%	7.2%	8.9%
AR/NR Turnover (days)	62	57	57	59
AP/NP Turnover (days)	56	55	54	55
Inventory Turnover (days)	47	46	45	46
Cash conversion cycle (days)	52	48	48	50

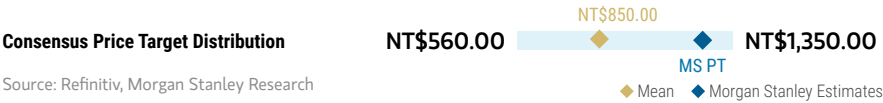
Source: Company data, Morgan Stanley Research (E) estimates.

Risk Reward – Unimicron (3037.TW)

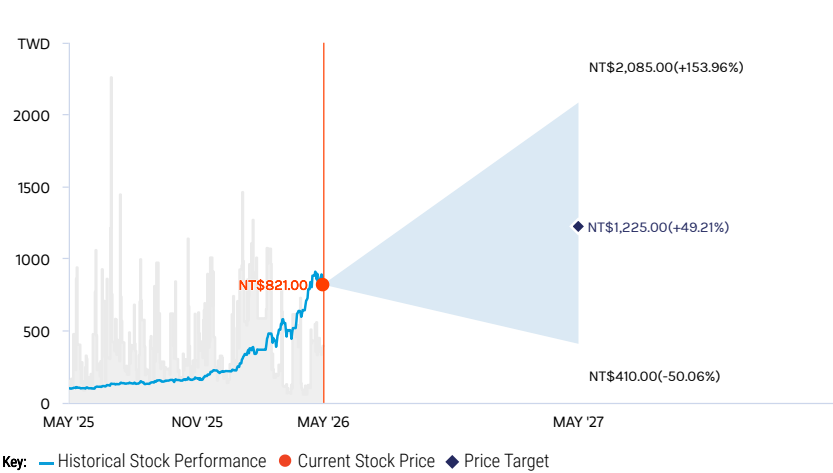
Price hikes and spec upgrades to support higher earnings estimates; OW

PRICE TARGET NT\$1,225.00

Base case, residual income (RI) valuation model, which we think derives the most accurate value of the firm given that it takes into account cost of equity. We use a cost of equity of 9.2% [risk-free rate of 1% (10-year Taiwan government note yield), equity risk premium of 8.7%, and a beta of 1.0], a medium-term growth rate of 15%, and a terminal growth rate of 3%.



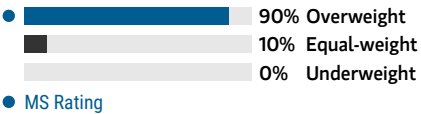
RISK REWARD CHART



OVERWEIGHT THESIS

- We expect the ABF substrate market to revert to undersupply from 2027 onwards, with the undersupply gap reaching 5-10% by 2030.
- This is primarily driven by AI demand, with next-gen AI/networking chips adopting larger body size ABF substrates.
- In the near to medium term, the short supply of T-glass will be a key watch point, but we believe this will mainly affect low-end ABF and BT substrates, as suppliers prioritize supply to the AI chip vendors.
- Our price target implies 51x 2027e P/E or 25x 2028e P/E, above the 10-20x range at which it traded in the previous upcycle. We deem this fair, given AI chip substrate spec upgrades and suppliers' ability to raise prices.

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Pricing Power: Positive

Secular Growth: Positive

View descriptions of Risk Rewards Themes [here](#)

BULL CASE NT\$2,085.00

87x 2027e P/E

Demand for AI servers, general servers and PCs is much stronger than expected: Pricing increases more than expected, 30%+ in 2026-27, driving stronger margin expansion for Unimicron.

BASE CASE NT\$1,225.00

51x 2027e P/E

Entering up-cycle driven by AI chip upgrades: Robust demand for AI GPUs and ASICs, supported by continued data center infrastructure investments by CSPs and neo cloud, leads to undersupply in the high-end ABF market from 2027e onwards. PC demand is sluggish, while general servers still show double-digit unit growth in 2026e. ASP has bottomed and started to inflect upwards, driven by raw material price hikes and substrate spec upgrades.

BEAR CASE NT\$410.00

17x 2027e P/E

Demand for AI servers and general servers weakens, with PC demand weakening further: The result is greater ABF oversupply; thus, there are pricing declines of ~15-20% in 2026-27, leading to weaker margins for Unimicron over the next two years.

Risk Reward – Unimicron (3037.TW)

KEY EARNINGS INPUTS

Drivers	2025	2026e	2027e	2028e
IC substrate ASP YoY (%)	7.5	30.5	32.6	29.3
IC substrate GM (%)	15.2	21.9	28.1	36.6

INVESTMENT DRIVERS

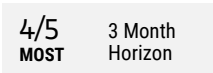
- 5G smartphone demand
- 5G base station demand
- Automotive demand

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies [here](#)

MS ALPHA MODELS



Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Better-than-expected ABF substrate demand from PC and server customers
- Capex cuts; halt to its capacity expansion plan
- Continued yield issues of alternative technology that doesn't require substrate (e.g., CoWoP)

RISKS TO DOWNSIDE

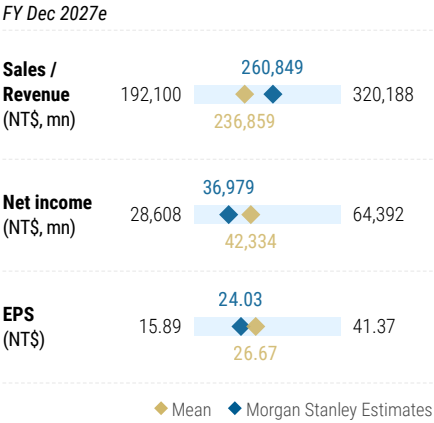
- Sudden demand shortfall
- Technological change that would not require ABF substrates
- Intensifying competition
- Yield issues or production hiccups when ramping new capacity

OWNERSHIP POSITIONING



Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS



Source: Refinitiv, Morgan Stanley Research

Risk Reward – Nan Ya PCB (8046.TW)

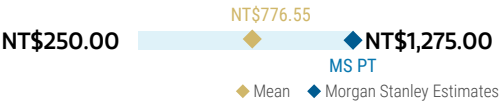
Price hikes and spec upgrades to support higher earnings estimates; OW

PRICE TARGET NT\$1,275.00

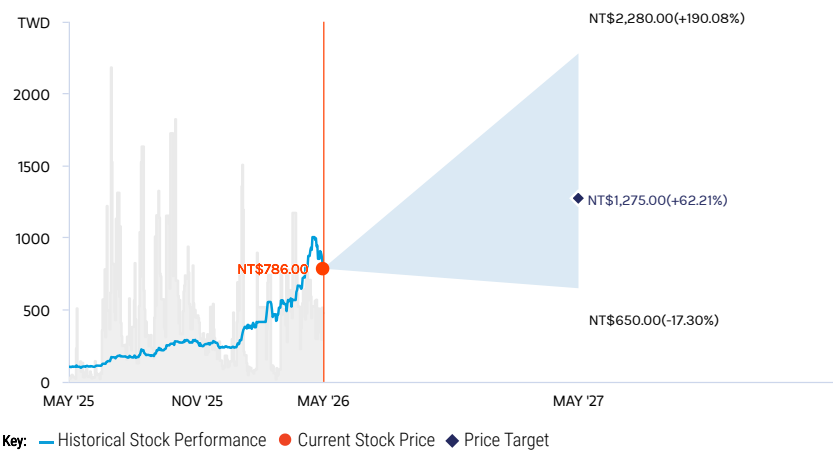
Base case, residual income (RI) valuation model, which we think derives the most accurate value of the firm given that it takes into account cost of equity. We use a cost of equity of 9.3% [risk-free rate of 1% (10-year Taiwan government note yield), equity risk premium of 8.7%, and a beta of 1.0], a medium-term growth rate of 17%, and a terminal growth rate of 3%.

Consensus Price Target Distribution

Source: Refinitiv, Morgan Stanley Research



RISK REWARD CHART

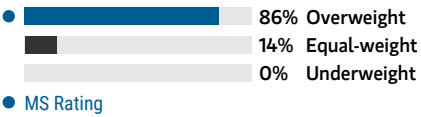


Source: Refinitiv, Morgan Stanley Research

OVERWEIGHT THESIS

- Price hikes for BT and ABF substrates, driven by raw material cost increases, will likely drive margin improvement for NYPCB.
- We believe the ABF substrate market will become undersupplied again from 2027 onwards, driven by AI demand, with next-gen AI/networking chips adopting large - body size ABF substrates.
- NYPCB should benefit from ABF substrate structural tailwinds as well - rising data traffic, big data analysis, AI/ML, and 5G.
- Our price target implies 54x 2027e P/E or 25x 2028e P/E, vs. the 10-30x range in which it traded in the previous upcycle. We deem this fair given ABF substrate spec upgrades and suppliers' ability to raise prices.

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Pricing Power: Positive
Secular Growth: Positive

View descriptions of Risk Rewards Themes [here](#)

BULL CASE

NT\$2,280.00

96x 2027e P/E

BT price increases further in 1H26, and ABF becomes undersupplied faster than expected (current expectation is from 2027e onwards), driving even stronger margin expansion for Nan Ya PCB.

BASE CASE

NT\$1,275.00

54x 2027e P/E

Entering up-cycle driven by AI chip upgrades: Robust demand for AI GPUs/ASICs and networking chips, supported by continued data center infrastructure investments by CSPs and neo cloud, leads to undersupply in the high-end ABF market from 2027e onwards. PC demand is sluggish, while general servers still show double-digit unit growth in 2026e. ASP has bottomed and started to inflect upwards, driven by raw material price hikes and substrate spec upgrades.

BEAR CASE

NT\$650.00

27x 2027e P/E

ABF demand weakens further, resulting in greater oversupply in the next few years. Thus, ABF pricing decreases 20% or more in 2026-27e, leading to weaker margins for Nan Ya PCB.

Risk Reward – Nan Ya PCB (8046.TW)

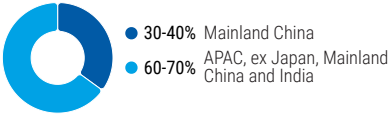
KEY EARNINGS INPUTS

Drivers	2025	2026e	2027e	2028e
ABF ASP y/y (%)	9.6	20.5	34.1	42.1
ABF GM (%)	15.1	23.1	32.3	44.0

INVESTMENT DRIVERS

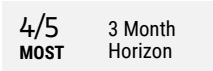
- 5G smartphone demand
- 5G base station demand
- Automotive demand

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies [here](#)

MS ALPHA MODELS



Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Better-than-expected ABF and BT demand
- Faster-than-expected pickup in AI and 5G demand
- Stronger-than-expected ASP hike
- Continued yield issues of alternative technology that doesn't require substrate (e.g. CoWoP)

RISKS TO DOWNSIDE

- A sudden demand shortfall, creating headwinds for ABF substrate demand and pricing
- Technological change that would not require ABF substrates
- Intensifying competition

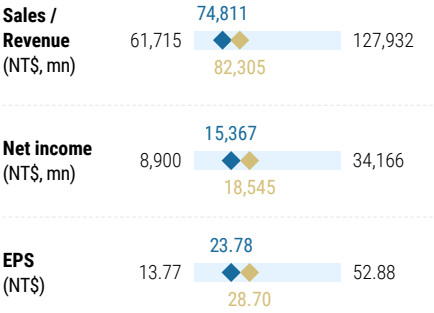
OWNERSHIP POSITIONING



Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS

FY Dec 2027e



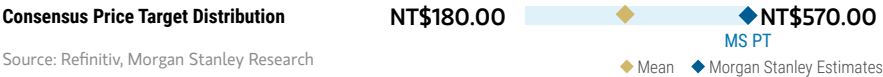
Source: Refinitiv, Morgan Stanley Research

Risk Reward – Zhen Ding (4958.TW)

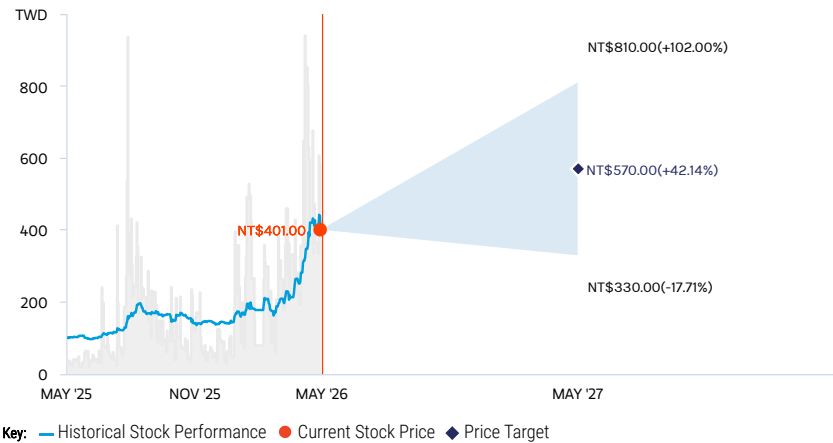
Growing China & International AI chip substrate and PCB vendor

PRICE TARGET NT\$570.00

Base case, residual income model. Key assumptions include a cost of equity of 10%, a medium-term growth rate of 15% and a terminal growth rate of 3%.



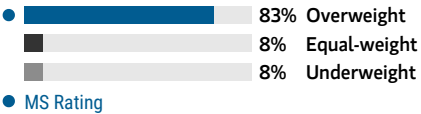
RISK REWARD CHART



OVERWEIGHT THESIS

- iPhone volumes are more resilient than Android given Apple's advantages on securing memory.
- ZDT is one of the key ABF substrate suppliers for China AI chips, which is expected to grow meaningfully over the next few years.
- ZDT is growing its ABF exposure to International clients as well, including Google TPU.
- Optical transceiver module PCB is a meaningful growth driver for ZDT, with very good margins.
- Shenzhen ABF fab 1 continues to ramp and has turned profitable in 1Q26, and fab 2 is now under construction. Kaohsiung fab is on track to enter mass production in 2H26.
- Our PT of NT\$570 implies 25x/19x CY27e/28e P/E.

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Pricing Power: *Negative*
View descriptions of Risk Rewards Themes [here](#)

BULL CASE	NT\$810.00	BASE CASE	NT\$570.00	BEAR CASE	NT\$330.00
35x 2027e P/E		25x 2027e P/E		14x 2027e P/E	
Greater share gains and better earnings contribution from new projects in ABF and AI Server PCBs: In the bull case, we assume ZDT is able to do Google TPU substrate extremely well and uses it to approach other ASIC/GPU customers, expanding its international client base. Additionally, we also assume ZDT expands Nvidia AI PCB share to 20-30% share on good execution and yield.		Stable demand for F-PCB and SLP, with share gains in BT/ABF substrates, and AI server PCBs: Zhen Ding is one of the main ABF substrate suppliers for China AI chips, while growing its share for international clients, potentially starting with Google TPU. At the same time, it continues to grow its AI server PCB share with Nvidia and other hyperscalers, and grows its exposure to optical module transceivers.		Weaker iPhone demand with share loss and ZDT does not execute on Google TPU substrates: iPhone demand misses market expectations, Zhen Ding loses share in F-PCB to Chinese peers, and SLP demand is also weak. We also assume here that ZDT fails to execute on TPU substrates and only continues to have exposure to China AI chips, with less than 5% share for Nvidia AI PCBs.	

Risk Reward – Zhen Ding (4958.TW)

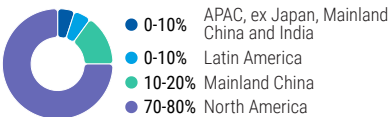
KEY EARNINGS INPUTS

Drivers	2025	2026e	2027e	2028e
F-PCB sales y/y (%)	5.0	4.1	13.7	7.3
F-PCB GM (%)	21.4	24.2	24.6	25.6
F-PCB ASP per sq ft (NT\$)	59.5	61.6	64.4	66.6

INVESTMENT DRIVERS

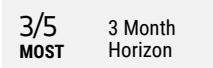
- iPhone, MacBook, iPad demand
- F-PCB content per iPhone, MacBook, iPad
- F-PCB ASP
- Yield rate for ABF/BT substrates

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies [here](#)

MS ALPHA MODELS



Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Higher-than-expected F-PCB content increase for iPhones
- Better-than-expected production yield/share allocation for SLP
- Share gains on higher margin F-PCB pieces

RISKS TO DOWNSIDE

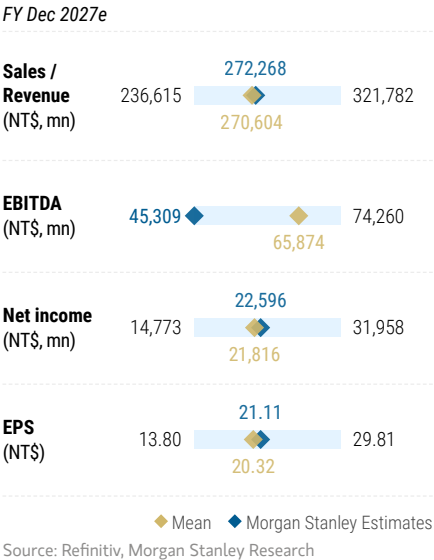
- Worse-than-expected iPhone sell-through
- Lower-than-expected F-PCB content increase for iPhones
- Worse-than-expected production yield/share allocation for SLP
- Increasing competition from Chinese peers, intensifying pricing pressure

OWNERSHIP POSITIONING



Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS



Valuation Methodology and Risks

Samsung Electro-Mechanics (009150.KS)

Base case, residual income valuation model. We apply a 10.5% cost of equity (beta 1.0, equity risk premium of 6.5%, and risk-free rate of 4%) and a 5% terminal growth rate. Our base year is 2026 and our forecast period is 2026-36.

Risks to Upside

- Price hikes driven by MLCC shortage
- Better-than-expected smartphone demand
- Higher MLCC margin from better yield and product mix
- Strong tailwind from China's consumption-friendly policies

Risks to Downside

- Significant pullback in Samsung's mobile flagship product cycle and SEMCO's lower market share
- Execution risk in penetrating Chinese smartphone customers
- Consumer demand headwinds in 2026

Ibiden (4062.T)

Derived from the base case. DCF assumptions: 2.2% risk-free rate, 1.31 beta, 3.7% risk premium, for 6.5% WACC. Terminal growth of 0% from F3/36.

Risks to Upside

- High-value-added ABF package products support earnings increase beyond our expectation from F3/27 2H.
- Sales exceed our base-case scenario as material shortages for FC packages are resolved

Risks to Downside

- Sales are weaker than base case because key material suppliers cannot supply enough material.
- We estimate OP sensitivity to FX changes at ¥800mn for ¥1/US\$ and ¥100mn for ¥1/€.

Risk Reward Reference links

1. View explanation of Options Probabilities methodology - [Options_Probabilities_Exhibit_Link.pdf](#)
2. View descriptions of Risk Rewards Themes - [RR_Themes_Exhibit_Link.pdf](#)
3. View explanation of regional hierarchies - [GEG_Exhibit_Link.pdf](#)
4. View explanation of Theme/Exposure methodology - [ESG_Sustainable_Solutions_External_Link.pdf](#)
5. View explanation of HERS methodology - [ESG_HERS_External_Link.pdf](#)

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Global Stock Ratings Distribution

(as of April 30, 2026)

The Stock Ratings described below apply to Morgan Stanley's Fundamental Equity Research and do not apply to Debt Research produced by the Firm.

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Stock Rating Category	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1546	42%	467	51%	30%	709	44%
Equal-weight/Hold	1568	43%	358	39%	23%	715	44%
Not-Rated/Hold	4	0%	0	0%	0%	1	0%
Underweight/Sell	555	15%	84	9%	15%	202	12%
Total	3,673		909			1627	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

Analyst Stock Ratings

Overweight (O). The stock's total return is expected to exceed the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Equal-weight (E). The stock's total return is expected to be in line with the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Not-Rated (NR). Currently the analyst does not have adequate conviction about the stock's total return relative to the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Underweight (U). The stock's total return is expected to be below the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

Analyst Industry Views

Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.

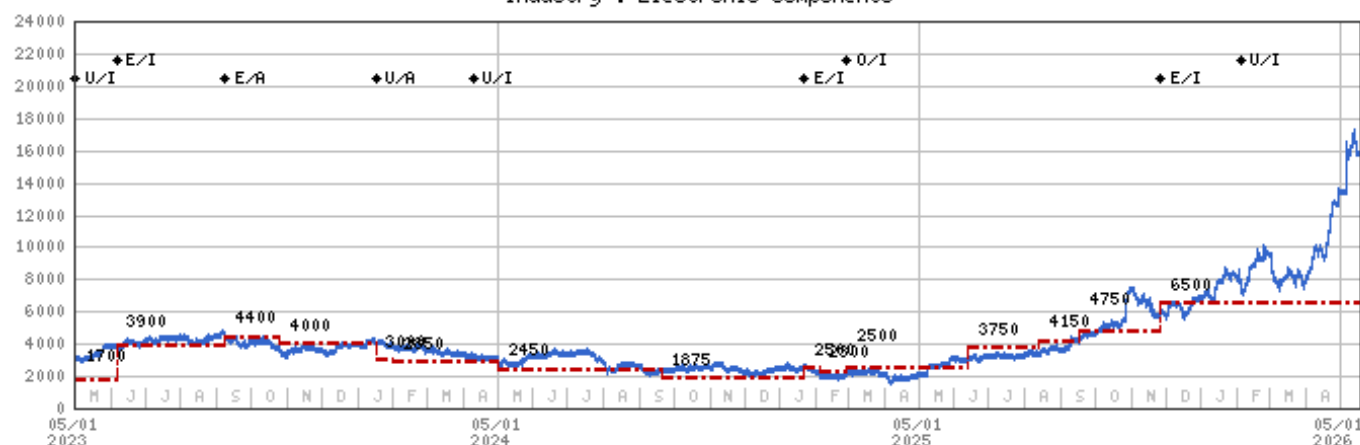
In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.

Cautious (C): The analyst views the performance of his or her industry coverage universe over the next 12-18 months with caution vs. the relevant broad market benchmark, as indicated below.

Benchmarks for each region are as follows: North America - S&P 500; Latin America - relevant MSCI country index or MSCI Latin America Index; Europe - MSCI Europe; Japan - TOPIX; Asia - relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Stock Price, Price Target and Rating History (See Rating Definitions)

Ibiden (4062.T) - As of 05/18/26 GMT in JPY
Industry : Electronic Components



Stock Rating History: 5/1/21 : O/I; 6/1/22 : U/I; 6/6/23 : E/I; 9/8/23 : E/A; 1/17/24 : U/A; 4/10/24 : U/I; 1/22/25 : E/I;
2/28/25 : O/I; 11/26/25 : E/I; 2/4/26 : U/I

Price Target History: 4/12/21 : 3200; 6/23/21 : 3650; 8/24/21 : 3750; 9/27/21 : 4000; 10/29/21 : 4200; 1/17/22 : 4250; 2/4/22 : 4100;
4/11/22 : 3750; 6/1/22 : 1875; 6/30/22 : 1550; 10/27/22 : 1750; 1/17/23 : 2000; 2/3/23 : 1750; 3/22/23 : 1700; 6/6/23 : 3900;
9/8/23 : 4400; 10/26/23 : 4000; 1/17/24 : 3000; 2/1/24 : 2850; 5/2/24 : 2450; 9/20/24 : 1875; 1/22/25 : 2500; 2/4/25 : 2300;
2/28/25 : 2500; 6/13/25 : 3750; 8/13/25 : 4150; 9/17/25 : 4750; 11/26/25 : 6500

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target — No Price Target Assigned (NA)

Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst) —

Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View

Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)

Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Nan Ya PCB (8046.TW) - As of 05/18/26 GMT in TWD
Industry : Greater China Technology Hardware



Stock Rating History: 5/1/21 : NA/I; 10/22/21 : O/I; 6/1/22 : E/I; 12/21/22 : U/I; 2/23/26 : O/I

Price Target History: 10/22/21 : 580; 11/9/21 : 630; 2/9/22 : 650; 2/25/22 : 670; 6/1/22 : 420; 8/23/22 : 275; 10/27/22 : 235;
12/21/22 : 180; 2/2/23 : 165; 2/25/23 : 155; 6/1/23 : 225; 11/7/23 : 170; 12/4/23 : 168; 1/4/24 : 160; 5/7/24 : 135; 8/9/24 : 112;
9/15/24 : 110; 10/8/24 : 100; 1/6/25 : 99; 4/15/25 : 70; 4/29/25 : 73; 7/1/25 : 75; 10/6/25 : 165; 2/23/26 : 515; 4/20/26 : 850

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target — No Price Target Assigned (NA)

Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst) —

Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View

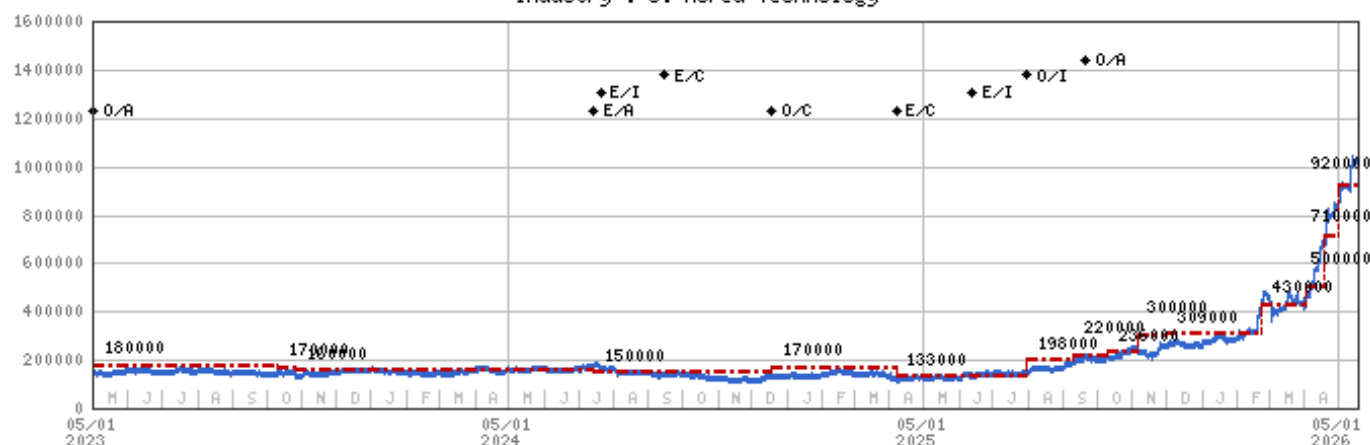
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)

Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Samsung Electro-Mechanics (009150.KS) - As of 05/18/26 GMT in KRW
Industry : S. Korea Technology



Stock Rating History: 5/1/21 : O/A; 5/12/21 : E/A; 7/19/21 : E/I; 8/12/21 : E/C; 6/1/22 : U/C; 10/4/22 : U/A; 11/9/22 : O/A; 7/15/24 : E/A; 7/21/24 : E/I; 9/15/24 : E/C; 12/18/24 : O/C; 4/7/25 : E/C; 6/13/25 : E/I; 7/31/25 : O/I; 9/21/25 : O/A

Price Target History: 3/1/21 : 250000; 5/12/21 : 180000; 10/19/21 : 170000; 10/27/21 : 160000; 1/24/22 : 180000; 3/18/22 : 170000; 6/1/22 : 140000; 11/9/22 : 180000; 10/11/23 : 170000; 10/27/23 : 160000; 7/15/24 : 150000; 12/18/24 : 170000; 4/7/25 : 133000; 7/31/25 : 198000; 9/10/25 : 220000; 10/10/25 : 236000; 11/5/25 : 300000; 12/1/25 : 309000; 2/23/26 : 430000; 4/2/26 : 500000; 4/19/26 : 710000; 5/1/26 : 920000

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target -- No Price Target Assigned (NA)

Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst) —

Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View

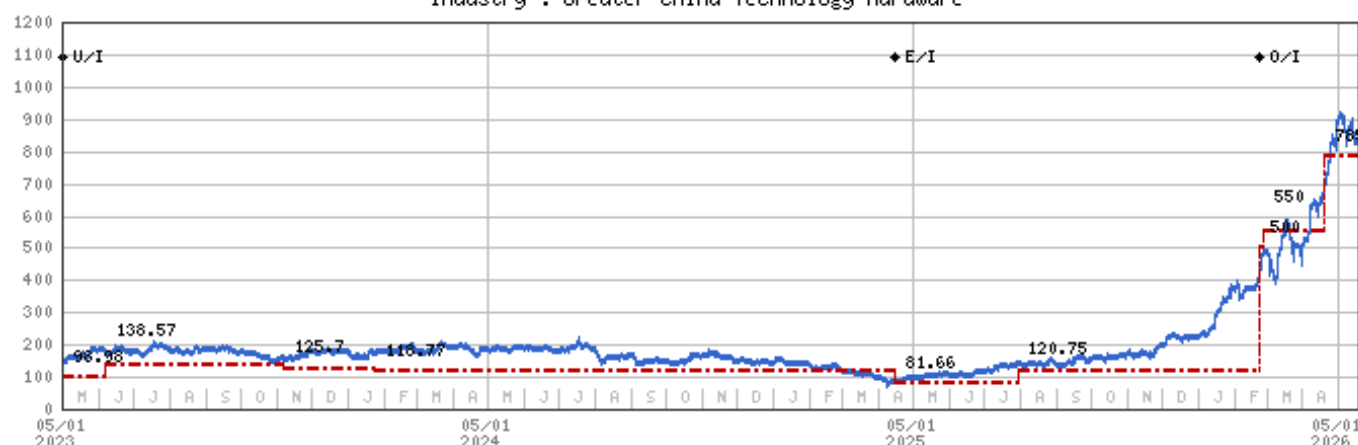
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)

Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

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Unimicron (3037.TW) - As of 05/18/26 GMT in TWD
Industry : Greater China Technology Hardware



Stock Rating History: 5/1/21 : NA/I; 10/22/21 : O/I; 6/1/22 : E/I; 2/22/23 : U/I; 4/15/25 : E/I; 2/23/26 : O/I

Price Target History: 10/22/21 : 188.06; 10/27/21 : 237.55; 2/9/22 : 267.24; 2/24/22 : 366.22; 6/1/22 : 237.55; 8/23/22 : 178.16; 10/27/22 : 133.62; 12/21/22 : 123.72; 2/2/23 : 116.79; 2/22/23 : 98.98; 6/7/23 : 138.57; 11/7/23 : 125.7; 1/24/24 : 118.77; 4/15/25 : 81.66; 7/30/25 : 120.75; 2/23/26 : 500; 2/26/26 : 550; 4/20/26 : 785

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target -- No Price Target Assigned (NA)

Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst) —

Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View

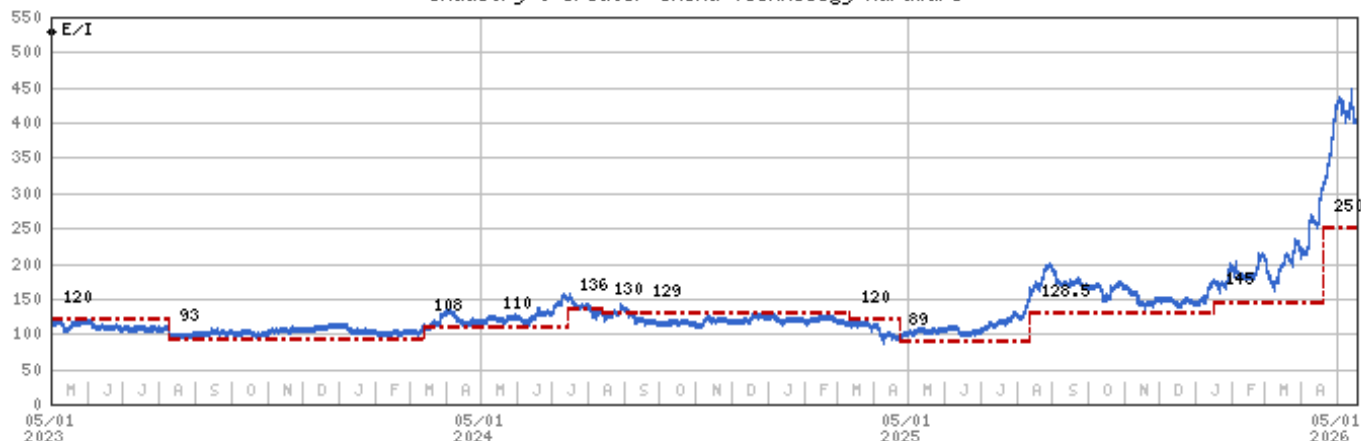
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)

Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Zhen Ding (4958.TW) - As of 05/18/26 GMT in TWD
Industry : Greater China Technology Hardware



Stock Rating History: 5/1/21 : U/I; 8/2/22 : E/I

Price Target History: 3/30/21 : 105; 8/2/22 : 124; 11/4/22 : 126; 3/14/23 : 120; 8/10/23 : 93; 3/13/24 : 108; 5/11/24 : 110; 7/15/24 : 136; 8/13/24 : 130; 9/15/24 : 129; 3/12/25 : 120; 4/23/25 : 89; 8/13/25 : 128.5; 1/16/26 : 145; 4/20/26 : 250

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target -- No Price Target Assigned (NA)

Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst)

Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View

Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)

Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

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INDUSTRY COVERAGE: Greater China Technology Hardware

COMPANY (TICKER)	RATING (AS OF)	PRICE* (05/15/2026)
Andy Meng, CFA		
AAC Technologies Holdings (2018.HK)	O (01/29/2024)	HK\$42.92
Accelink Technologies Co. Ltd. (002281.SZ)	U (05/12/2022)	Rmb196.49
BYD Electronics (0285.HK)	O (04/28/2023)	HK\$27.26
China TransInfo Technology Co Ltd (002373.SZ)	E (07/18/2023)	Rmb8.57
Dahua Technology Co. Ltd. (002236.SZ)	E (12/12/2024)	Rmb17.50
Eoptolink Technology Inc Ltd (300502.SZ)	O (02/27/2026)	Rmb610.05
Genius Electronic Optical Co. Ltd. (3406.TW)	E (04/23/2025)	NT\$588.00
Gosuncn Technology Group Co Ltd (300098.SZ)	U (11/07/2022)	Rmb5.90
HIKVision Digital Technology (002415.SZ)	E (12/12/2024)	Rmb33.12
Largan Precision (3008.TW)	E (10/17/2025)	NT\$3,445.00
LianChuang Electronic Technology Co Ltd (002036.SZ)	U (06/12/2024)	Rmb8.38
OFILM Group Co Ltd (002456.SZ)	U (06/12/2024)	Rmb8.75
Q Technology (Group) Company Ltd (1478.HK)	U (03/23/2026)	HK\$8.64
Quectel Wireless Solutions Co Ltd (603236.SS)	E (10/09/2024)	Rmb73.09
Shanghai Conant Optical Co Ltd (2276.HK)	O (04/10/2026)	HK\$44.18
Shenzhen Transsion Holdings Co Ltd (688036.SS)	E (03/23/2026)	Rmb60.70
Sunny Optical (2382.HK)	E (03/23/2026)	HK\$64.95
Suzhou TFC Optical Communication Co Ltd. (300394.SZ)	E (11/06/2025)	Rmb399.99
Wingtech Technology Co Ltd (600745.SS)	E (11/10/2023)	Rmb18.66
Xiaomi Corp (1810.HK)	O (04/14/2021)	HK\$30.70
Yangtze Optical Fibre and Cable JSC Ltd (601869.SS)	U (10/13/2021)	Rmb374.45
Yangtze Optical Fibre and Cable JSC Ltd (6869.HK)	E (04/20/2023)	HK\$223.00
Yongxin Optics Co Ltd (603297.SS)	E (11/15/2022)	Rmb140.45
Zhejiang Crystal-Optech Co Ltd (002273.SZ)	O (11/15/2022)	Rmb38.00
Zhongji Innolight Co Ltd (300308.SZ)	++	Rmb1,049.87
ZTE Corporation (0763.HK)	E (03/11/2024)	HK\$25.22
ZTE Corporation (000063.SZ)	U (07/02/2021)	Rmb36.71

Derrick Yang

Accton Technology Corporation (2345.TW)	O (06/06/2024)	NT\$2,505.00
Advantech (2395.TW)	O (01/20/2021)	NT\$459.50
AirTAC International (1590.TW)	O (04/16/2025)	NT\$1,465.00
AU Optronics (2409.TW)	E (02/10/2026)	NT\$19.05
Bizlink (3665.TW)	O (03/10/2025)	NT\$2,095.00
BOE Technology (000725.SZ)	O (09/06/2019)	Rmb4.14
Chenbro (8210.TW)	O (07/23/2025)	NT\$1,410.00
Chroma Ate Inc. (2360.TW)	O (10/05/2021)	NT\$2,240.00
E Ink Holdings Inc. (8069.TWO)	O (05/11/2026)	NT\$210.00
Ennostar Inc (3714.TW)	U (09/23/2022)	NT\$67.20
Hiwin Technologies Corp. (2049.TW)	O (03/30/2026)	NT\$392.50
Innolux (3481.TW)	E (04/07/2025)	NT\$36.10
King Slide Works Co. Ltd. (2059.TW)	O (11/08/2023)	NT\$4,860.00
Lens Technology (300433.SZ)	E (07/22/2020)	Rmb34.21
Radiant Opto-Electronics Corporation (6176.TW)	E (03/01/2024)	NT\$99.60
Sanan Optoelectronics (600703.SS)	U (08/21/2023)	Rmb16.16
TCL Corp. (000100.SZ)	E (04/07/2025)	Rmb4.18
Tianma Microelectronics (000050.SZ)	U (01/24/2018)	Rmb7.47
Wuhan Jingce Electronic Group Co Ltd (300567.SZ)	E (11/26/2021)	Rmb175.00

Howard Kao

Acer Inc. (2353.TW)	U (04/23/2025)	NT\$27.70
Asustek Computer Inc. (2357.TW)	U (11/16/2025)	NT\$641.00
Compal Electronics (2324.TW)	U (04/23/2025)	NT\$28.65
FIT Hon Teng Ltd (6088.HK)	O (11/03/2025)	HK\$9.39
Giga-Byte Technology Co. Ltd. (2376.TW)	E (11/16/2025)	NT\$326.50
Gold Circuit Electronics Ltd. (2368.TW)	O (10/06/2022)	NT\$1,290.00
Inspur Electronic Information (000977.SZ)	E (08/28/2023)	Rmb70.21
Lenovo (0992.HK)	E (11/16/2025)	HK\$12.42
Lotes Co. Ltd. (3533.TW)	E (05/12/2025)	NT\$2,330.00
Nan Ya PCB (8046.TW)	O (02/23/2026)	NT\$786.00
Pegatron Corporation (4938.TW)	E (03/25/2026)	NT\$76.50
Quanta Computer Inc. (2382.TW)	O (05/01/2023)	NT\$306.00
Shengyi Technology Co Ltd. (600183.SS)	E (05/26/2022)	Rmb89.95
Shennan Circuits Co Ltd (002916.SZ)	E (08/24/2023)	Rmb323.25
Unimicron (3037.TW)	O (02/23/2026)	NT\$821.00
Wistron Corporation (3231.TW)	O (07/12/2023)	NT\$135.50
Wiwynn Corp (6669.TW)	O (11/10/2025)	NT\$5,115.00
Yageo Corp. (2327.TW)	O (10/28/2025)	NT\$455.50
Zhen Ding (4958.TW)	O (05/18/2026)	NT\$401.00

Sharon Shih

Asia Vital Components Co. Ltd. (3017.TW)	O (07/30/2024)	NT\$2,455.00
Auras Technology Co Ltd (3324.TWO)	E (05/04/2023)	NT\$995.00
Catcher Technology (2474.TW)	U (11/17/2025)	NT\$185.00
Delta Electronics Inc. (2308.TW)	O (07/13/2017)	NT\$2,075.00
Fositek Corp (6805.TW)	O (06/25/2025)	NT\$1,735.00
Foxconn Industrial Internet Co. Ltd. (601138.SS)	O (07/10/2019)	Rmb67.62
Foxconn Technology (2354.TW)	U (04/23/2025)	NT\$59.40
GoerTek Inc (002241.SZ)	U (04/23/2025)	Rmb25.33
Hon Hai Precision (2317.TW)	O (03/15/2024)	NT\$248.50
LandMark Optoelectronics Corporation (3081.TWO)	E (03/26/2026)	NT\$2,655.00
Lingyi Itech Guangdong Co (002600.SZ)	U (04/23/2025)	Rmb16.94
Lite-On Technology (2301.TW)	E (01/15/2025)	NT\$204.50
Luxshare Precision Industry Co., Ltd. (002475.SZ)	O (10/24/2016)	Rmb74.00
Sunonwealth Electric Machine Industry Co (2421.TW)	E (02/23/2024)	NT\$140.00

Tong Hsing (6271.TW)	E (03/18/2019)	NT\$177.00
Visual Photonics Epitaxy Co Ltd (2455.TW)	E (09/11/2023)	NT\$358.50

Stock Ratings are subject to change. Please see latest research for each company.
* Historical prices are not split adjusted.

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